Chapter 1

What is Numerical Analysis about ?

Case 1. Minimizer

Theorem 1. *Suppose $f : [a, b] \to \mathbb{R}$ is continuous on $[a, b]$. Let $m := \inf\{f(x) \mid x \in [a, b]\}$. Then there exists a minimizer $x_* \in [a, b]$ such that $f(x_*) = m$.*

Note: we have used that a continuous function f on $[a, b]$ is bounded.

Proof. Step 1. For each $n = 1, 2, 3, \cdots$ let $x_n \in [a, b]$ be any element such that

$$f(x_n) < m + \frac{1}{n}.$$

For each n, such an element must exist: Otherwise, i.e., if $f(x) \geq m + \frac{1}{n}$ for every $x \in [a, b]$, then $m + \frac{1}{n}$ is a lower bound of the range set that is strictly bigger than m, contradicting to the definition of m.

Step 2. We have defined the sequence (x_n), and let $y_n := f(x_n)$. Then $y_n \to m$ as $n \to \infty$ because

$$0 \leq |y_n - m| < \frac{1}{n}.$$

Step 3. (x_n) is a sequence of real numbers in $[a, b]$, and thus (x_n) is a bounded sequence. Therefore, there exists a convergent subsequence $(x_{n_k})_{k=1}^{\infty}$ to a limit x_*. Furthermore, because $[a, b]$ is closed, the limit point $x_* \in [a, b]$. We also know that the subsequence (y_{n_k}) of a convergent sequence (y_n) also is convergent to the same limit m.

Step 4. Now, we observe

$$m = \lim_{k\to\infty} y_{n_k} = \lim_{k\to\infty} f(x_{n_k}) = f(x_*),$$

where in the last equality, we used the fact that f is continuous on $[a, b]$. $\square$

Case 2 : rank of a matrix

Let $A \in \mathbb{R}^{n\times k}$ be a real $n \times k$ matrix. The very first fundamental notion for A is the **rank of** A, the number being **defined** by the fact that

the number of linearly independent columns = the number of linearly independent rows.

Once the assertion of the equality is proved, the definition of the rank will make sense.

Denote the $(LHS) = r_{col}(A)$ and $(RHS) = r_{row}(A)$. This fundamental equality is a consequence of the inequality that

$$\text{for any matrix } A, \quad r_{row}(A) \le r_{col}(A)$$

because if that is true then $\Big(r_{col}(A) = \Big)r_{row}(A^T) \le r_{col}(A^T)\Big(= r_{row}(A)\Big)$.

The inequality follows from the assertion

if $r_{i_1}, r_{i_2}, \cdots r_{i_\rho}$ are linearly independent rows of A
then $Ar_{i_1}, Ar_{i_2}, \cdots Ar_{i_\rho}$ are linearly independent vectors in $\mathbb{R}^n$.

Here we denote the rows of A from top to bottom by r_1, r_2, $\cdots$, r_n. We examine the proof of the assertion.

Proof. Suppose not, i.e., we have $(\lambda_1, \lambda_2, \cdots, \lambda_\rho) \neq 0$ such that $\sum_{\alpha=1}^{\rho} \lambda_\alpha\big(Ar_{i_\alpha}\big) = 0$. Then,

$$0 = \sum_{i=\alpha}^{\rho} \lambda_\alpha (Ar_{i_\alpha}) = \sum_{\alpha=1}^{\rho} A(\lambda_\alpha r_{i_\alpha}) = A\Big(\sum_{\alpha=1}^{\rho} \lambda_\alpha r_{i_\alpha}\Big).$$

But,

$$Ax = 0 \quad \Longleftrightarrow \quad x \perp \mathsf{span}\,\big\langle r_1, r_2, \cdots, r_n\big\rangle.$$

Combining the two observations,

$$\sum_{\alpha=1}^{\rho} \lambda_\alpha r_{i_\alpha} \in \mathsf{span}\,\big\langle r_1, r_2, \cdots, r_n\big\rangle \cap \mathsf{span}\,\big\langle r_1, r_2, \cdots, r_n\big\rangle^{\perp} = \{0\}.$$

Thus, we conclude that $\sum_{\alpha=1}^{\rho} \lambda_\alpha r_{i_\alpha}$ is a zero vector. Recalling that $r_{i_1}, r_{i_2}, \cdots r_{i_\rho}$ are linearly independent, we conclude $\lambda_1 = \lambda_2 = \cdots = \lambda_\rho = 0$. Contradiction. $\square$

This is the way we are taught in Linear Algebra course: we prove the equality, and based on this fact, we give the definition for the rank. However, this proof does not give a clue on how do we know the rank of A.

This second example is the case where, there exists another proof that contains a clue for the calculation.

Singular Value Decomposition

Theorem 2. *For any matrix $A \in \mathbb{R}^{n \times k}$, then there exist*

1. *an $n \times n$ orthogonal matrix O,*
2. *a $k \times k$ orthogonal matrix Q,*
3. *an $n \times k$ matrix Λ where nonzero entries are only at diagonals Λ_{ii}, $i = 1, 2, \cdots, min(n, k)$*

so that $A = O\Lambda Q^T$. The number of nonzero entries of Λ coincides with the rank of A.

I will not provide the proof of the Singular Value Decomposition, but its proof contains a way to compute O, Q, and Λ, and this provides the mean to compute the rank.

Therefore, in some cases, there can be two proofs:

1. one that is optimal for the proof, easy to read, maybe short, etc.
2. the other that is optimal for the implementaion, maybe longer, etc.

so the material for the Numerical Analysis may be taught from different point of view.

Case 3 : Riemann Integral or Quadrature

The third example is the case, where the definition contains a clue for a computation.

Let us consider a high-school series calculations. Let n be a fixed natural number. For each i, we let $x_i = a + i\frac{b-a}{n}$. We let t_i be any point belogs to $[x_{i-1}, x_i]$ for $i = 1, \cdots n$.

The definition of the Riemann Integral of a continuous function f over $[a, b]$ is justified by **the existence of a real number** I such that

$$\lim_{n\to\infty} \sum_{i=1}^{n} f(t_i)\ \frac{b-a}{n} = I.$$

regardless of choice (t_i) at each n. We examine the proof of the assertion.

Proof. Step 1. By extreme value theorem on each compact interval $[x_{i-1}, x_i]$, there exist the minimizer t_{i*} and t_i^* in $[x_{i-1}, x_i]$. Then we have for each n

$$\sum_{i=1}^{n} f(t_{i*})\ \frac{b-a}{n} \le \sum_{i=1}^{n} f(t_i)\ \frac{b-a}{n} \le \sum_{i=1}^{n} f(t_i^*)\ \frac{b-a}{n}.$$

Step 2. We omit the proof that $\left(\sum_{i=1}^{n} f(t_{i*})\ \frac{b-a}{n}\right)_{n=1}^{\infty}$ and $\left(\sum_{i=1}^{n} f(t_i^*)\ \frac{b-a}{n}\right)_{n=1}^{\infty}$ are bounded monotone sequences so that their respective limits $\alpha, \beta \in \mathbb{R}$, $\alpha \le \beta$ exist.

Step 3. That f is continuous in $[a, b]$ implies the following:

$$\begin{aligned}&\text{For every } \epsilon > 0, \text{ there exists } \delta > 0 \text{ such that}\\ &x, x' \in [a, b], |x - x'| < \delta \quad \Longrightarrow \quad |f(x) - f(x')| < \frac{\epsilon}{b-a}.\end{aligned}$$

that is to say, f is uniformly continuous in $[a, b]$. For a given ϵ, we take $K \in \mathbb{N}$ so that $\frac{b-a}{K} < \delta$ for each δ. This gives that

for every ϵ, there exists K such that for $n \ge K$,

$$\Big|\sum_{i=1}^{n} f(t_i^*)\ \frac{b-a}{n} - \sum_{i=1}^{n} f(t_{i_*})\ \frac{b-a}{n}\Big| \le \sum_{i=1}^{n} \Big|f(t_i^*) - f(t_{i*})\Big|\ \frac{b-a}{n} \le \sum_{i=1}^{n} \frac{\epsilon}{b-a}\ \frac{b-a}{n} = \epsilon.$$

This implies that the limits $\alpha = \beta$. By the Squeeze theorem, $\left(\sum_{i=1}^{n} f(t_i)\ \frac{b-a}{n}\right)_{n=1}^{\infty}$ is convergent, to the same limit. The limit is defiend to be I. $\square$

Therefore in some cases,

1. the proof we are taught contains a well-working method.
2. Since there is no reason we stick to this method only, of course we explore quite much better working numerical methods (midpoint rule, simpson rule, gaussian quadrature, $\cdots$) than the one suggested.

Case 4 : Jordan factorization of a square matrix T.

Another striking example where we do not have a general algorithm stably working for it is the Jordan factorization.

Theorem 3. *Let A be an $n \times n$ real square matrix. Then there exist*

1. *an $n \times n$ invertible matrix P,*
2. *an $n \times n$ matrix J in real Jordan form,*

such that $A = PJP^{-1}$.

Try to re-visit its proof in Linear Algebra textbook. They only assert the existence. More precisely, the steps that possibly compute P and J contains the precedure whose small perturbation gives rise to completely different result.

What will we treat in this course ?

1. Definition of a sequence (a_n), and a function $f : [a, b] \to \mathbb{R}$.
2. interpolating / approximating polynomials for a class of functions. Topics: spline, Tchebyshev polynomials, $\cdots$.
3. Existence of fixed point and Newton's method on nonlinear system.
4. Existence of solution of initial value problem of nonlinear ode / Runge-Kutta methods, $\cdots$
5. For all of above, we will conduct the error analysis.
6. (2nd semester) Linear algebra theory or higher dimensional techniques and more on (partial) differential equations.

Chapter 2

How does my computer throw $\sin(x)$ for an input x?

Case 1. A function from $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ to $\mathbb{R}$

We can define a function $a : \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\} \mapsto \mathbb{R}$, using the table.

n	1	2	3	4	5	6	7	8	9	10
a_n										

Case 2. A function from $\mathbb{N}$ to $\mathbb{R}$

Are there ways to define a function from countably infinite set to $\mathbb{R}$?

Case 3. A function from $[0, 1] \subset \mathbb{R}$ to $\mathbb{R}$

Are there ways to define a function from this uncountably infinite set $[0, 1]$ to $\mathbb{R}$?

A function on $E \subset \mathbb{R}^n$?

Is a function a black box ?
Is a function a table ?
Can you come up with an example of a function that you can build as a machine ?
How does my computer know which value to return for $\sin(x)$?

A few observations

1. A recursive definition for a sequence
   - For a sequence (a_n) it is possible to have a recursive definition.
   - A recursive definition makes sense on the set $\mathbb{N}$, and in general on the set with *well-ordering.*
   - $[a, b]$ *with usual ordering* $<$ *is not the case.*

2. The use of exponential b^n
   - For a given number b, b^n with n integer can be computed by finite use of arithmetics.
   - This is again a privilege of sequence which is not given to the function:

     $$\text{Let} \quad f(x) = b^x, \quad x \in \mathbb{R},\ b > 0.$$

     in case x is irrational number, b^x cannot be done in the same way as before.

3. For the use of $id(x) : x \mapsto x$, one sees that
   - The return value is computed <u>on the fly</u> at the time x is decided.
   - Hence, what we are doing is not like making a table, but more resembles the *pinball* machine.
   - As its consequence, in reality this device will handle only the rational input number x.

It seems, there are not many others than the class of elementary pp functions.

I wanted you to see the importance of a class of functions on $\mathbb{R}^n$ that are

(1) piecewise polynomial, and

(2) each piece's domain (the support more precisely) is elementary in the sense that you can finish the membership check of a given x to the domain in a finite procedures.

We will call those functions an elementary pp function.

Examples of sets $E \subset \mathbb{R}^n$, where the membership check is finitely done.

Remark 2.1. The literal definition of the function on an uncountable set, which might suggest the tabularized mapping, indeed is beyond our intuition.

Remark 2.2. For functions that are continuous or measurable, the relation between elemetary pp functions and them are like the relation between rational numbers and real numbers.

A continuous function on $\mathbb{R}^n$ is **a limit of continuous elementary piecewise polynomial functions**, where the convergence is uniform in every compact subset of $\mathbb{R}^n$.

A measurable function on $\mathbb{R}^n$ is **a limit of elementary piecewise polynomial functions**, where the convergence is pointwise, for almost everywhere $x \in \mathbb{R}^n$.

Therefore, in principle, a suitable quantity can be computed as accurate as we want. This yields the concept of computing the limit and error control.

Objective of our study

1. We may assume sufficiently but finitely many sampling data for a function f:

$$\big(x_i, f(x_i)\big)_{i=1,\cdots,N}.$$

2. For this sample data, we come up with a pp function φ such that

$$\varphi(x_i) = f(x_i), \quad i = 1, 2, \cdots, N.$$

3. Out of
   (a) The sample data,
   (b) The assumption that f was, say m times continuously differentiable.

   we look for the remainder formula for other values of x

$$R(x) = f(x) - \varphi(x).$$

Questions

1. How do we choose a pp function φ? $\rightarrow$ theory and method
2. How accurate is it ? $\rightarrow$ The remainder formula.

One piece vs. manu pieces

For these two questions,

First, we will take one piece, out of piecewise polynomial function, and seek for the remainder formular, restricted on that piece's support. (Local study)

Second, we assemble pieces into one pp function. (Global study)

Chapter 3

One Piece: (Local) Polynomial Interpolation

Later, we will consider a pp function on $[a_0, b_0]$, where the domain is divided into many pieces by

$$a_0 = t_0 < t_1 < t_2 < \cdots < t_L = b_0, \quad I_j = [t_{j-1}, t_j], \quad j = 1, 2, \cdots, L.$$

For today's discussion, we pick one of the piece $I = I_j$, and proceed the local problem in that one piece.

Let $I = [a, b]$. Fix an integer $n \geq 0$.

Sample Data

With repect to the integer $n \geq 0$, we choose, or we are given, the $n+1$ sampling points

$$x_1, x_2, \cdots, x_{n+1} \in [a, b].$$

We will denote the choice of $n+1$ points by

$$[x_1, x_2, \cdots, x_{n+1}].$$

The choice $[x_1, x_2, \cdots, x_{n+1}]$ then will assume in our class:

If x_i is repeated $m+1$ times $\quad f(x_i), f'(x_i), \cdots, f^{(m)}(x_i) \quad$ are collected at x_i,

where f is sufficiently many times differentiable.

Sets of functions

We consider

$$C^{n+1}([a,b]) = \big\{ f : [a,b] \to \mathbb{R} \mid f \text{ is } n+1 \text{ times differentiable and } f,\, f',\, f'',\, \cdots,\, f^{(n+1)} \text{ are continuous in } [a,b] \big\}.$$

Note: At the endpoint a and b, the differentibility is the right-differentiability and the left-differentiability respectively.

We also let

$$\mathcal{P}^{<n+1}([a,b]) = \text{the set of polynomials of order at most } n \text{ restricted on } [a,b].$$

Interpolation problem

For a given choice,

1. We look for a polynomial $p \in \mathcal{P}^{<n+1}([a,b])$ that matches the data, for example,

   $$p(x_i) = f(x_i), \quad p'(x_i) = f'(x_i), \quad \cdots, \quad p^{(m)}(x_i) = f^{(m)}(x_i).$$

   For f and choice of points, we will say another function p matches the data if the collected data of p and f coincide in the above sense.

2. We seek for a formula of the remainder $R(x)$,

   $$R(x) = f(x) - p(x), \quad x \in [a,b]$$

In fact, the term *interpolation* does not necessarily assume the function f and does not assume that the data are collected by sampling. Nevertheless, it is convenient to speak of the function f.

For those who major in this field, the classes of functions we can treat similarly are

$$\begin{aligned} C^{n+1}([a,b]) \subset O^{n+1}([a,b]) &= \big\{ f : [a,b] \to \mathbb{R} \mid (n+1)\text{-th distributional derivative of } f \text{ is in } L^1([a,b]) \big\} \\ \subset \tilde{O}^{n+1}([a,b]) &= \big\{ f : [a,b] \to \mathbb{R} \mid (n+1)\text{-th distributional derivative of } f \text{ is a Radon measure.} \big\} \end{aligned}$$

Fundamental Theorem of Calculus, Taylor's Theorem

We recall first the fundamental theorem of calculus:

Theorem 1. *Let $f \in C^1([a,b])$ and $x_1 \in [a,b]$. Then for $x \in [a,b]$,*

$$R(x) = f(x) - f(x_1) = \int_{x_1}^{x} \frac{df}{dx}(x')\, dx' = (x - x_1) \int_{\tau=0}^{1} \frac{df}{dx}\big((1-\tau)x_1 + \tau x\big)\, d\tau.$$

Mean Value Theorem: there exists a $\xi \in [a,b]$ such that

$$R(x) = f(x) - f(x_1) = (x - x_1) f'(\xi).$$

Theorem 2 (Taylor's theorem)**.** *Let $f \in C^{n+1}([a,b])$ and $x_1 \in [a,b]$. Then for $x \in [a,b]$,*

$$\begin{aligned} R(x) &= f(x) - \Big(f(x_1) + f'(x_1)(x - x_1) + \frac{f''(x_1)}{2!}(x - x_1)^2 + \cdots + \frac{f^{(n)}(x_1)}{n!}(x - x_1)^n\Big) \\ &= (x - x_1)^{n+1} \times \frac{1}{n!} \int_{\tau=0}^{1} (1-\tau)^n \frac{d^{n+1} f}{dx^{n+1}}\Big((1-\tau)x_1 + \tau x\Big)\, d\tau. \end{aligned}$$

Mean Value like Theorem: there exists a $\xi \in [a,b]$ such that

$$R(x) = \frac{(x - x_1)^{n+1}}{(n+1)!} f^{(n+1)}(\xi).$$

Remark 3.1. We take the FTC as the $n = 0$ case of the Taylor's theorem. We take the Taylor's therem as the case where the choice of points and data were all at x_1,

$$x_1, x_1, x_1, \cdots, x_1 \quad (n+1) \text{ times.}$$

We will prove the similar remainder theorem for the arbitrary choice of the $n+1$ points.

The notation $\big([x_1, x_2, \cdots, x_{\ell+1}], f\big)$

Let $I = [a, b]$, $n \geq 0$ an integer, $\underline{f \in C^{n+1}([a, b])}$. Let $0 \leq \ell \leq n+1$ be an integer and we consider $\ell + 1$ points

$$x_1, x_2, x_3, \cdots, x_{\ell+1} \in [a, b] \quad \text{(may be repeated)}.$$

A choice $[x_1, \cdots, x_{\ell+1}]$ acts on the class $\underline{C^{n+1}([a, b])}$, denoted by

$$\big([x_1, x_2, \cdots, x_{\ell+1}], f\big) \in \mathbb{R}.$$

For the sake of mathematical clarity, this action is defined by nontrivial surface integrals in the next page. Its motivational uses are given first.

Properties of the action $\big([x_1, x_2, \cdots, x_{\ell+1}], f\big)$

(1) The value is independent of orders in the choice $[x_1, x_2, \cdots, x_{\ell+1}]$.

(2) Example of $\ell = 1$ case:

$$\begin{aligned} &\text{if } x_1 \neq x_2, && \big([x_1, x_2], f\big) = \frac{f(x_2) - f(x_1)}{x_2 - x_1} = \frac{f(x_1) - f(x_2)}{x_1 - x_2}, \\ &\text{if } x_1 = x_2 = \bar{x}, && \big([\bar{x}, \bar{x}], f\big) = f'(\bar{x}). \end{aligned}$$

(3) For $1 \leq \ell \leq n+1$, the equality always holds:

$$\big([x_1, x_2, \cdots, x_{\ell+1}], f\big)\ (x_{\ell+1} - x_1) = \big([x_2, x_3, \cdots, x_{\ell+1}], f\big) - \big([x_1, x_2, \cdots, x_\ell], f\big).$$

Indeed, above $\ell = 1$ calculations verify

$$\big([x_1, x_2], f\big)(x_2 - x_1) = \big([x_2], f\big) - \big([x_1], f\big).$$

Definition

Definition 1. *For $f \in C^{n+1}([a,b])$, and $0 \le \ell \le n+1$ integer,*

$$\begin{aligned} &\text{if } \ell = 0 && \big([x_1], f\big) = f(x_1), \\ &\text{if } \ell \ge 1 && \big([x_1, x_2, \cdots, x_{\ell+1}], f\big) := \frac{1}{\sqrt{\ell+1}} \int_{\Lambda^\ell \subset \mathbb{R}^{\ell+1}} \frac{d^\ell f}{dx^\ell}\big(\lambda_1 x_1 + \cdots \lambda_{\ell+1} x_{\ell+1}\big)\, dS(\lambda). \end{aligned}$$

The set Λ^ℓ is the regular ℓ-simplex

$$= \Big\{(\lambda_1, \lambda_2, \cdots, \lambda_{\ell+1}) \in \mathbb{R}^{\ell+1} \mid \forall e \quad \lambda_e \ge 0, \quad \sum_{e=1}^{\ell+1} \lambda_e = 1\Big\} \subset \mathbb{R}^{\ell+1}$$

The integral is denoted by $\big([x_1, x_2, \cdots, x_{\ell+1}], f\big)$, or $[x_1, \cdots, x_{\ell+1}]f$, or $f[x_1, x_2, \cdots, x_{\ell+1}]$.

1d remainder theorem

Having defined the actions for $f \in C^{n+1}([a,b])$, we can state the remainder theorem.

Theorem (1d remainder theorem)**.** *Let $f \in C^{n+1}([a,b])$ and x_1, x_2, $\cdots$, $x_{n+1} \in [a,b]$. Then, for $x \in [a,b]$,*

$$\begin{aligned} R(x) = f(x) - \Big(&\big([x_1], f\big) + \big([x_1, x_2], f\big)(x - x_1) + \big([x_1, x_2, x_3], f\big)(x - x_1)(x - x_2) + \cdots \\ &+ \big([x_1, x_2, \cdots, x_{n+1}], f\big)(x - x_1)(x - x_2)(x - x_3)\cdots(x - x_n)\Big) \\ &= (x - x_1)(x - x_2)\cdots(x - x_n)(x - x_{n+1})\big([x_1, x_2, x_3, \cdots, x_{n+1}, x], f\big). \end{aligned}$$

Remark 3.2. .

1. Note that the remainder is in the form of integral, that is the definition $\big([x_1, x_2, x_3, \cdots, x_{n+1}, x], f\big)$.
2. We will also give its *mean value evaluation* too.

$$\exists \xi \in [a,b] \quad \big([x_1, x_2, x_3, \cdots, x_{n+1}, x], f\big) = \frac{1}{(n+1)!} f^{(n+1)}(\xi).$$

3. Taylor's theorem is a special case where $x_1 = x_2 = \cdots = x_{n+1}$.

Proof of 1d remainder theorem

All we need is the key equality

$$f[x_1, x_2, \cdots, x_{\ell+1}]\ (x_{\ell+1} - x_1) = f[x_2, x_3, \cdots, x_{\ell+1}] - f[x_1, x_2, \cdots, x_\ell]. \qquad \text{(KEY)}$$

We will give a proof of this Key equality, and mean value evaluation as well, later using the surface integral definition.

proof of 1d remainder theorem. For $k \in \{0, 1, 2, \cdots, n\}$, we let

$$\begin{aligned} p_{<k+1}(x) = \Big(&\big([x_1], f\big) + \big([x_1, x_2], f\big)(x - x_1) + \big([x_1, x_2, x_3], f\big)(x - x_1)(x - x_2) + \cdots \\ &+ \big([x_1, x_2, \cdots, x_{k+1}], f\big)(x - x_1)(x - x_2)(x - x_3) \cdots (x - x_k)\Big). \end{aligned}$$

We assert that for $k \in \{0, 1, 2, \cdots, n\}$,

$$f(x) - p_{<k+1}(x) = \big([x_1, x_2, \cdots, x_{k+1}, x], f\big)(x - x_1)(x - x_2)(x - x_3) \cdots (x - x_{k+1}).$$

At $k = 0$, by the key equality

$$f(x) - p_{<1}(x) = f(x) - f(x_1) = \big([x_1, x], f\big)(x - x_1).$$

Now, if the assertion is true for $0, 1, 2, \cdots, k - 1$,

$$\begin{aligned} f(x) - p_{<k+1}(x) &= f(x) - p_{<k}(x) - f[x_1, x_2, \cdots, x_{k+1}](x - x_1)(x - x_2)(x - x_3) \cdots (x - x_k) \\ &= f[x_1, x_2, \cdots, x_k, x](x - x_1)(x - x_2)(x - x_3) \cdots (x - x_k) \\ &\quad - f[x_1, x_2, \cdots, x_k, x_{k+1}](x - x_1)(x - x_2)(x - x_3) \cdots (x - x_k) \\ &= f[x_1, x_2, \cdots, x_k, x_{k+1}, x]\Big((x - x_0)(x - x_1)(x - x_2) \cdots (x - x_{k-1})(x - x_{k+1})\Big). \end{aligned}$$

□

A few instances of surface integral calculation:

$\ell = 1$ **case**:

1. From the definition
$$\big([x_1, x_2], f\big) = \frac{1}{\sqrt{2}} \int_{\Lambda^1 \subset \mathbb{R}^2} \frac{df}{dx}\big(\lambda_1 x_1 + \lambda_2 x_2\big)\, dS(\lambda) = J,$$
where Λ^1 is the line segment $\{(\lambda_1, \lambda_2) \in \mathbb{R}^2 \mid \lambda_1, \lambda_2 \geq 0, \lambda_1 + \lambda_2 = 1\}$.

2. We let $\tau \in [0,1]$ and consider a map $[0,1] \to \Lambda^1$, $\tau \mapsto (\lambda_1(\tau), \lambda_2(\tau)) = (1-\tau, \tau)$. By change of variable formula,
$$J = \frac{1}{\sqrt{2}} \int_{[0,1]} \frac{df}{dx}\big(\lambda_1(\tau) x_1 + \lambda_2(\tau) x_2\big) \sqrt{D\lambda^T D\lambda}\, d\tau,$$
where $D\lambda$ is the derivative matrix $\begin{pmatrix} \frac{d\lambda_1}{d\tau} \\ \frac{d\lambda_2}{d\tau} \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$. Hence,
$$J = \int_{[0,1]} \frac{df}{dx}\big((1-\tau)x_1 + \tau x_2\big)\, d\tau.$$

3. If $x_1 = x_2$, then the integral
$$J = \int_{[0,1]} \frac{df}{dx}(x_1)\, d\tau = \frac{df}{dx}(x_1) \int_{[0,1]} d\tau = \frac{df}{dx}(x_1)$$
as expected.

4. If $x_1 \neq x_2$, let $\tilde{f}(\tau) := f\big((1-\tau)x_1 + \tau x_2\big)$, $\quad \frac{\partial \tilde{f}}{\partial \tau}(\tau) = \frac{\partial f}{\partial x}\big((1-\tau)x_1 + \tau x_2\big)(x_2 - x_1)$.
$$J = \int_{[0,1]} \frac{1}{x_2 - x_1} \frac{\partial \tilde{f}}{\partial \tau}(\tau)\, d\tau = \frac{\tilde{f}(1) - \tilde{f}(0)}{x_2 - x_1} = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$
as expected.

$\ell = 2$ **case,** $x_1 = x_2 = x_3$:

1. From the definition
$$\big([x_1, x_1, x_1], f\big) = \frac{1}{\sqrt{3}} \int_{\Lambda^2 \subset \mathbb{R}^3} \frac{d^2 f}{dx^2}\big(\lambda_1 x_1 + \lambda_2 x_1 + \lambda_3 x_1\big)\, dS(\lambda) = J,$$
where Λ^2 is the regular triangle $\{(\lambda_1, \lambda_2, \lambda^3) \in \mathbb{R}^3 \mid \lambda_1, \lambda_2, \lambda_3 \geq 0, \lambda_1 + \lambda_2 + \lambda_3 = 1\}$.

2. In the special case of $x_1 = x_2 = x_3$,
$$J = \frac{d^2 f}{dx^2}(x_1) \frac{1}{\sqrt{3}} \int_{\Lambda^2 \subset \mathbb{R}^3} dS(\lambda).$$

3. We may directly compute the surface area of the regular triangle: Since the side length is $\sqrt{2}$, the area is $\frac{\sqrt{3}}{2}$. Hence,
$$J = \frac{1}{2} \frac{d^2 f}{dx^2}(x_1).$$

We postpone the full general calculations until the end of this chapter.

Existence and Uniqueness of interpolating polynomial

Theorem 1 (uniqueness)**.** *For any $f \in C^{n+1}([a,b])$ and points $x_1, x_2, \cdots, x_{n+1} \in [a,b]$, the interpolating polynomial in $\mathcal{P}^{<n+1}([a,b])$ is unique.*

Proof. Suppose the sampling points were such that

$$\text{distinct} \quad x_1, x_2, \cdots, x_N \quad \text{whose multiplicities add up } n+1.$$

Suppose $p, q \in \mathcal{P}^{<n+1}([a,b])$ be the two interpolating polynomials. Then $r = p - q \in \mathcal{P}^{<n+1}([a,b])$ too, i.e., order at most n. The polynomial r have roots whose total multiplicities are $n+1$. We conclude $r \equiv 0$. □

Theorem 2 (existence)**.** *For any $f \in C^{n+1}([a,b])$ and points $x_1, x_2, \cdots, x_{n+1} \in [a,b]$, there exists a polynomial $p \in \mathcal{P}^{<n+1}([a,b])$ that matches the data.*

We will give a proof, first for the simple case where $x_1, x_2, \cdots, x_{n+1}$ are all distinct, second for the case $n+1$ is an even number and every points are precisely 2 times repeated, and lastly for general case.

case 1: $x_1, x_2, \cdots, x_{n+1}$ are all distinct. Recall the remainder

$$f(x) - p_{<n+1}(x) = \big([x_1, x_2, \cdots, x_{n+1}, x], f\big)(x - x_1)(x - x_1)\cdots(x - x_{n+1}),$$

Since $(x - x_1)(x - x_1)\cdots(x - x_{n+1})$ will be zero at every sampling points, we only need to check if $\big([x_1, x_2, \cdots, x_{n+1}, x], f\big)$ can diverge or not. By definition

$$\big([x_1, x_2, \cdots, x_{n+1}, x], f\big) = \frac{1}{\sqrt{n+2}} \int_{\Lambda^{n+1} \subset \mathbb{R}^{n+2}} \frac{d^{n+1} f}{dx^{n+1}} \big(\lambda_1 x_1 + \cdots \lambda_{n+1} x_{n+1} + \lambda_{n+2} x\big)\, dS(\lambda),$$

and since $x \mapsto \frac{d^{n+1} f}{dx^{n+1}} \big(\lambda_1 x_1 + \cdots \lambda_{n+1} x_{n+1} + \lambda_{n+2} x\big)$ is a continuous function,

$$x \mapsto \big([x_1, x_2, \cdots, x_{n+1}, x], f\big)$$

is a continuous function in $[a,b]$, and its range is bounded. □

case 2: $x_1, x_2, \cdots, x_{2L}$, every point is 2 times repeated. This is the case $n+1 = 2L$. Recall the remainder

$$R(x) = f(x) - p_{<2L}(x) = \big([x_1, x_1, x_2, x_2, \cdots, x_L, x_L, x], f\big)(x - x_1)^2 (x - x_2)^2 \cdots (x - x_L)^2.$$

For similar reason in the case 1, we check

$$f(x_i) - p_{<2L}(x_i) = 0, \quad i = 1, 2, \cdots, L.$$

We let $x \mapsto \big([x_1, x_1, x_2, x_2, \cdots, x_L, x_L, x], f\big)$ be a function $g(x)$, which is continuous in $[a,b]$. While we know $R \in C^{n+1}([a,b])$, we compute the derivative at x_i directly. We write

$$R(x) = \tilde{g}(x)(x - x_i)^2, \quad \tilde{g} \text{ continuous in } [a,b].$$

$$\frac{R(x_i + h) - R(x_i)}{h} = \frac{R(x_i + h)}{h} = \frac{h^2 \tilde{g}(x_i + h)}{h} = h\tilde{g}(x_1 + h) \to 0 \quad \text{as } h \to 0.$$

□

general case proof is in fact the same argument as in the second case, making use of that (i): $R \in C^{n+1}([a,b])$ and that (ii):

$$\big([x_i, x_i+h, x_i+2h, \cdots, x_i+mh], R\big) \to \big([x_i, x_i, x_i, \cdots, x_i], R\big) = \frac{1}{m!} R^{(m)}(x_i).$$

general case. Suppose x_i is repeated $m+1 \geq 1$ times. Claim: $R(x_i) = R'(x_i) = \cdots = R^{(m)}(x_i) = 0$.
Proof of the claim: We write

$$R(x) = w(x)(x-x_i)^{m+1}, \quad \text{with } w \text{ continuous.}$$

For $\alpha = 0, 1, \cdots, m$,

$$\big([x_i, x_i+h, x_i+2h, x_i+\alpha h], R\big)$$

is precisely a combination of terms of the form $w(x_i+\beta h)\beta^{m+1}h^{m+1}$ divided by a scalar multiple of h^α, which tends to 0 as $h \to 0$.
Hence,

$$R^{(\alpha)}(x_i) = 0, \quad \alpha = 0, 1, \cdots, m.$$

$\square$

Chapter 4

Arithmetics of approximate real numbers

What we will do (by CPU) will be basically,

1. A few real numbers are stored in Computer.
2. We carry out a few calculations of $+, -, \times, /$.
3. Computer will return a value to me.

Therefore, we attempt to understand each of three steps.

1/ Floating number

- Nowadays, a number in our computer is processed in such a way, 64bit signal at once.
- 64 bits consist of
$$\pm a_1 a_2 \cdots a_m 2^{\pm e_1 e_2 \cdots e_t}$$
  1. 1 sign bit of the number,
  2. nonnegative binary integer $a_1 a_2 \cdots a_m$, for some $m < 64$,
  3. 1 sign bit of the exponent,
  4. nonnegative binary integer $e_1 e_2 \cdots e_t$, for some $t < 64$

  totally consuming 64bits.
- In this course, we will call such a 64bit tuple, <u>a floating number</u>.

Consequences of representing real number by floating number:

1. We have the limited maximum number of significant (binary) digits that is m, and any real number will be truncated by round off whose binary expansion is infinitely long for instances:

   irrational number π,

   The number $\dfrac{1}{5}$

   which is expanded finitetly in decimal representation but not in binary representation.

2. There is the largest positive number M:

$$a_1 = a_2 = \cdots = a_m = 1, \quad e_1 = e_2 = \cdots e_t = 1 \quad \text{two sign bits are both } +.$$

   Similarly, there is the minimum negative number. Hence, any value exceeding this range cannot be processed by CPU.

   We say, an intermediate floating number calculation raises *overflow* if the value exceeds the range.

3. There is the closest positive number δ to 0:

$$a_1 = a_2 = \cdots = a_{m-1} = 0, \quad a_m = 1, \quad \text{the exponent sign is negative}, \quad e_1 = e_2 = \cdots = e_t = 1.$$

   Hence, any value whose absolute value is less than δ may be technically stored as zero floting number.

   We say, an intermediate floating number calculation raises *underflow* if the absolute value is less than δ.

2/ Arithmetics of approximate numbers

In general, the input numbers we have in mind to store, usually also are supposed to have their own numbers of significant digits, for example, when we consider a quadratic polynomial

$$ax^2 + bx + c,$$

we may have in mind a, b, c are those numbers say, we are confident in amount having 5 siginificant digits in Decemial representation, with scale range $10^0 \sim 10^2$, for instance.

The following is about arithmetics of approximate numbers, and are independent of the discussion above on the floating number.

We examine two nonzero positive approximate numbers

$$x = 0.0023, \quad y = 230.0$$

We observe for x:

1. It has 2 siginficant digits.
2. Its scientific notation is

$$2.3 \times 10^{-3}.$$

We observe for y:

1. It has 4 siginficant digits.
2. Its scientific notation is

$$2.300 \times 10^2.$$

Out of the scientific notation, we can extract information:

1. The number scale is revealed in the exponent:

$$x \text{ is a number of order } 10^{-3} \text{ and } y \text{ is of order } 10^2.$$

2. The possible error from the true value can be understood in the following way.

   - Suppose the true value $\hat{x}$ was 0.00225. Then the error is 0.00005.

     $$0.00005 = \left(5 \times 10^{-2}\right)\left(10^{-3}\right)$$

     $$\text{(error scale)} = \text{(relative error scale)} \times \text{(number scale)}.$$

     where the relative error scale is decided by the siginificant digits 2.
   - Similarly, suppose the true value $\hat{y}$ was 230.04 Then the error is 0.04.

     $$0.04 = \left(4 \times 10^{-4}\right)\left(10^{2}\right)$$

     $$\text{(error scale)} = \text{(relative error scale)} \times \text{(number scale)}.$$

     where the relative error scale is decided by the siginificant digits 4.

3. In summary,

   - the number of siginificant digits decides the relative error scale,
   - and relative error scale times number scale is the error scale.

We write two nonzero positive approximate numbers a and b as

$$a = \hat{a}(1+\eta_a), \quad b = \hat{b}(1+\eta_b) \quad \hat{a},\, \hat{b} \text{ repective true values.}$$

Related to the earlier discussion,

$$\begin{aligned} |\eta_a| &\simeq \text{relative error scale decided by significant digits of } a \text{ expression,} \\ |\hat{a}| &\simeq \text{number scale of } a, \\ |\hat{a}||\eta_a| &\simeq \text{error scale of } a. \end{aligned}$$

Similar for b follows.

Below, we assume say, a has more siginificant digits than b, i.e.,

$$|\eta_a| \ll |\eta_b| \ll 1.$$

1. Addition

$$a + b = \hat{a} + \hat{a}\eta_a + \hat{b} + \hat{b}\eta_b = \big(\hat{a}+\hat{b}\big)\big(1 + \frac{\hat{a}}{\hat{a}+\hat{b}}\eta_a + \frac{\hat{b}}{\hat{a}+\hat{b}}\eta_b\big)$$

   - relative error scale

     $$|\eta_{a+b}| = \left|\frac{\hat{a}}{\hat{a}+\hat{b}}\eta_a + \frac{\hat{b}}{\hat{a}+\hat{b}}\eta_b\right| \leq |\eta_a| + |\eta_b| \simeq |\eta_b| \quad \text{since } |\eta_a| \ll |\eta_b| \text{ by assumption.}$$

   - number scale

     $$|\hat{a}+\hat{b}| \quad \text{follows one of the larger one of } |\hat{a}| \text{ and } |\hat{b}|$$

2. Multiplication

$$ab = \hat{a}(1+\eta_a)\hat{b}(1+\eta_b) = (\hat{a}\hat{b})\big(1+\eta_a+\eta_b+\eta_a\eta_b\big).$$

   - relative error scale

$$|\eta_{ab}| \le |\eta_a| + |\eta_b| + |\eta_a\eta_b| \simeq |\eta_b| \quad \text{since } |\eta_a|, |\eta_a\eta_b| \ll |\eta_b| \text{ by assumption.}$$

   - number scale is the new scale decided by

$$|\hat{a}\hat{b}|$$

3. Division

$$\frac{a}{b} = \frac{\hat{a}(1+\eta_a)}{\hat{b}(1+\eta_b)} = \frac{\hat{a}}{\hat{b}}\big(1+\eta_a\big)\big(1-\eta_b+C\eta_b^2\big) \quad \text{for some } C.$$

   - relative error scale

$$|\eta_{a/b}| \le |\eta_a| + |\eta_b| + \text{higher order terms} \simeq |\eta_b|$$

   - number scale is the new scale decided by

$$\left|\frac{\hat{a}}{\hat{b}}\right|$$

4. Subtraction

$$a-b = \hat{a}+\hat{a}\eta_a-\hat{b}-\hat{b}\eta_b = \big(\hat{a}-\hat{b}\big)\big(1+\frac{\hat{a}}{\hat{a}-\hat{b}}\eta_a-\frac{\hat{b}}{\hat{a}-\hat{b}}\eta_b\big)$$

   - relative error scale is now affected by the scale of

$$\frac{\hat{a}}{\hat{a}-\hat{b}} \quad \text{and} \quad \frac{\hat{b}}{\hat{a}-\hat{b}}.$$

     Suppose it happened to be $a = \hat{a} = 2.3001, \quad b = \hat{b} = 2.3$ Then

$$\left|\frac{\hat{a}}{\hat{a}-\hat{b}}\right| \simeq 10^4$$

     and thus the relative error scale will be

$$\simeq 10^4 \times |\eta_b| \simeq 10^4 \times 10^{-2} = 10^2 \quad !$$

   - number scale is the scale decided by

if $\hat{a}$ and $\hat{b}$ are not close,	$\left|\hat{a}-\hat{b}\right|$	follows one of the larger scale.
if $\hat{a} \simeq \hat{b}$,	$\left|\hat{a}-\hat{b}\right|$	follows the new small scale.

Summary:

1. The relative error scale follows the worse case (many places $|\eta_b|$ above), and the resultant siginicant digits follows that of b there.
2. Among all 4 arithmetics of nonzero positive approximate numbers, the only calculation where the relative error scale possibly become strictly worse, i.e., we lost siginificant digits, is the subtraction.
3. The error scale will be in general of <u>new scale:</u>

   (the relative error scale) times (the resultant (new) number scale) = (the resultant (new) error scale)

3/ Confidence on the computer's returning value

Typically,

1. My input numbers are such that the number of significant digits are known.
2. 64bits are long enough so that those approximate numbers are stored without losing any of digits.
3. Suppose calculation I ran did not reduce the siginificant digits. Then, In assessing the computer-returned 64bit, I wll be confident up to the a priori known number of significant digits.
4. Suppose calculation I ran reduced the digits, by for example a subtraction of two close numbers. I will be only confident
   up to the reduced number of significant digits and I should throw out the rest bits.

A list of risky codes in scientific computations

We examined three stages of

1. A few real numbers are stored in Computer.
2. We carry out a few calculations of $+, -, \times, /$.
3. Computer will return a value to me.

When we write a code, which will be calculating floating numbers, the discussion above leads to the following conclusion:

1. Don't Do:

   Do not make expressions, possibly leads to the loss of siginificant digits, i.e., subtracting two close numbers. For example, in calculating two roots of quadratic polynomial,

   $$x_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}, \quad x_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

   has a subtraction of two close number in the numerator in case c is very small. Specifically, if $b > 0$ x_1 has $\simeq -b + b$, and if $b < 0$ x_2 has $\simeq -b - |b|$.

   We should write a code, if $b > 0$,

   $$x_1 = \frac{2c}{-b - \sqrt{b^2 - 4ac}}, \quad x_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

   and if $b < 0$,

   $$x_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}, \quad x_2 = \frac{2c}{-b + \sqrt{b^2 - 4ac}}$$

2. There are at least three "Be careful":

   (i) Division by a small number.

   In principle, this does not reduce the siginifiant digits. However, if we have a different mind from what really happens, then results will not be as our anticipation. Because of the very large <u>number scale</u>, error scale (not relative error scale) will be larger than our wrongful anticipation.
   For example, suppose we write a code without having in mind that a matrix A may be close to be singular, using expression

   $$A^{-1}x.$$

   If we happen to input A that is close to a singular matrix, then this will not make the siginicant digits reduced, but the result error scale will be out of our wrongful anticipation.

   (ii) overflow

   If we deal with very large scale floating numbers, and if we are not sufficiently careful, our code might stop running: not because of expression has bad calculation or code has bug, but because overflow happens.

   To prevent this, use appropriate normalization of floating numbers, and let calculations be in reasonable number scales.

   (iii) underflow

   During the calculation, a number that is very close to 0 might have been replaced by 0 but we might not notice it if we are not careful.

   For example, many writes a code dividing cases

   if (float expression) $== 0$: case 1 jobs to do
   else : case 2 jobs to do

   What will really happens may be

   if $|$(float expression)$| < \delta$: case 1 jobs to do
   else : case 2 jobs to do.

   In other words, if we are not careful so that we think some nonzero expression would fall into the case 2, but it may fall into the case 1.

   To avoid the confusion, write the code in the latter style:

   $$\text{what happens} = \text{what's written.}$$

Chapter 5

Methods for the Lagrange Interpolation

Case 1. Data $\big(x_1, f(x_1)\big), \cdots, \big(x_{n+1}, f(x_{n+1})\big)$, x_i all distinct.

1. The same polynomial $p \in \mathcal{P}^{<n+1}([a,b])$ of order at most n can be expanded in many different ways. For instance, the polynomial

$$\begin{aligned} p_{<n+1}(x) = f[x_1] + f[x_1, x_2](x - x_1) + f[x_1, x_2, x_3](x - x_1)(x - x_2) + \cdots \\ + f[x_1, x_2, \cdots, x_{n+1}](x - x_1)(x - x_2)(x - x_3) \cdots (x - x_n), \end{aligned}$$

   which is expanded with polynomials of increasing orders with respect to the sampling point x_i, of course can be re-arranged in the form

$$p_{<n+1}(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n.$$

2. The former is said to be expanded in Newton basis, while the latter is said to be expanded in power basis. Newton basis and power basis both <u>are bad</u>.

Remark 5.1. Although the newton basis expansion of p is bad, at least one can compute it because the coefficient $f[x_1, x_2, \cdots, x_{\ell+1}]$ can be computed by the key recursive equality: because $x_1, x_2, \cdots, x_{\ell+1}$ are distinct

$$\begin{aligned} f[x_1] &= f(x_1) \\ f[x_1, x_2] &= \frac{f(x_2) - f(x_1)}{x_2 - x_1} \\ f[x_1, x_2, x_3] &= \frac{\frac{f(x_3)-f(x_2)}{x_3-x_2} - \frac{f(x_2)-f(x_1)}{x_2-x_1}}{x_3 - x_1} \\ &\vdots \end{aligned}$$

Remark 5.2. A basis what is called Lagrange basis and another basis what is called the Bernstein basis are in general said to better behave. Here we implement polynomial construction expanded in Lagrange basis.

Definition 1. *Let $x_1, x_2, \cdots, x_{n+1} \in [a, b]$ be distinct points. We define for each $i = 1, \cdots, n+1$,*

$$L_i(x) = \beta_i(x - x_1)(x - x_2)\cdots(x - x_{i-1})(x - x_{i+1})\cdots(x - x_{n+1}) \in \mathcal{P}^{<n+1}([a, b]),$$

where $\beta_i = \Big((x_i - x_1)(x_i - x_2)\cdots(x_i - x_{i-1})(x_i - x_{i+1})\cdots(x_i - x_{n+1})\Big)^{-1} = (\gamma_i)^{-1}$.

A few observations:

1. By assumption that $x_1, x_2, \cdots, x_{n+1}$ are distinct, $\gamma_i \neq 0$ and thus β_i is well-defined.
2. $L_i(x_i) = 1$ and $L_i(x_j) = 0$ if $i \neq j$.
3. For every i, the order of $L_i(x)$ is precisely n.

Remark 5.3. While $L_i(x)$ is of order n for every i, for a combination $\sum_{i=0}^{n} f(x_i)L_i(x)$ terms may cancel so that the resultant polynomial is of order less than n.

Proposition 2. *The interpolating polynomial $p \in \mathcal{P}^{<n+1}([a, b])$ obtained uniquely by Theorem 1,2 in the preceding section must be*

$$\sum_{i=0}^{n} f(x_i)L_i(x).$$

Proof. We check that $q(x) = \sum_{i=1}^{n+1} f(x_i)L_i(x)$ is a member of $\mathcal{P}^{<n+1}([a, b])$, and satisfies

$$q(x_i) = f(x_i) \quad i = 1, 2, \cdots, n+1.$$

Hence q is an interpolating polynomial in $\mathcal{P}^{<n+1}([a, b])$. Since the interpolating polynomial is unique, q coincides with the one obtained by Theorem 1,2. $\square$

Here, we devise two implementations of the function that returns the value

$$q(x) : x \mapsto \sum_{i=1}^{n+1} f(x_i)L_i(x).$$

The First implementation

We consider the literal implementation of the right-hand-side of $q(\bar{x})$.

1. We assume the sampling points $x_s = (x_1, x_2, \cdots, x_{n+1})$ and $f_s = (f(x_1), f(x_2), \cdots, f(x_{n+1}))$ are given as $n+1$ size vectors.
2. Recalling that

   $$L_i(x) = \beta_i(x - x_1)(x - x_2)\cdots(x - x_{i-1})(x - x_{i+1})\cdots(x - x_{n+1}),$$

   we literally compute

   $$\beta_i = \Big(x_i - x_1)(x_i - x_2)\cdots(x_i - x_{i-1})(x_i - x_{i+1})\cdots(x_i - x_{n+1})\Big)^{-1}$$
   $$m_i(x) = (x - x_1)(x - x_2)\cdots(x - x_{i-1})(x - x_{i+1})\cdots(x - x_{n+1})$$

   for $\bar{x}$ in place of x.

   (blank)

3. We literally compute the combination $\sum_{i=1}^{n+1} f(x_i)\beta_i m_i(\bar{x})$.

   (blank)

4. This works perfectly fine.

Remark 5.4. Note that in the linear combination $\sum_{i=1}^{n+1} f(x_i)\beta_i m_i(\bar{x})$, it possibly conducts subtraction of two close numbers.
Let us set our confidence level on the sampling data to be $\eta = 10^{-7}$, or data were with 7,8 siginificant digits.

1. Suppose the returned value $|q(\bar{x})| \geq \eta$ then no problem, we have a similar confidence on returned value.
2. If $|q(\bar{x})| < \eta$, no matter the returned value was

   $$0.000000015, \text{ or } 0.0000000185, \text{ or, } 10^{-22}$$

   they are all negelected and we safely is confident on that $|q(\bar{x})| < \eta$.

The second Implementation

Once a certain method stably works, one now seeks efficiency or other factors, provided that the new method works equally good.

1. In computing $L_i(\bar{x}) = \beta_i m_i(\bar{x})$, it is tempting we compute $m_i(\bar{x})$ not making n products, but making one division with

   $$w(\bar{x}) = (\bar{x}-x_1)(\bar{x}-x_2)\cdots(\bar{x}-x_{n+1}). \quad \text{If } \bar{x} \notin \{x_1, x_2, \cdots, x_{n+1}\}, \quad L_i(\bar{x}) = \beta_i \frac{w(\bar{x})}{\bar{x}-x_i},$$

   $w(\bar{x})$ is computed once and is shared afterwards.

   This is to say that the first implementation is an $O(n^2)$ algorithm while the second implementation is an $O(n)$ algorithm.

2. The expression $\beta_i \frac{w(\bar{x})}{\bar{x}-x_i}$ is the case of "Be careful" when $\bar{x} \neq x_i$ but very close to x_i. We justify why this is safe:

   Let the numerator $h(x) = \beta_i w(x)$. Then, we can see

   $$\begin{aligned} h(x) &= h(x_i) + h'(x_i)(x - x_i) + \text{high order terms in } (x - x_i) \\ h(x_i) &= 0, \quad h'(x_i) = 1. \end{aligned}$$

   This tells that, near x_i,

   - The numerator and the denominator will be of very small but same scale $|x - x_i|$,
   - The siginificant digits will not be lost in multiplication and division,
   - Thus, $L(\bar{x})$ will be obtained with same significant digits, the number scale is 10^0, and the error scale is as same as the given relative error scale.

Having known the confidence of the expression, in fact, since there is another quite smart way, we introduce further idea.

1. Here we continue to use the expression but we arrange $q(\bar{x})$ in the form

   $$q(\bar{x}) = \begin{cases} f(x_i) & \text{if } \bar{x} \in \{x_1, x_2, \cdots, x_{n+1}\} \text{ and } \bar{x} = x_i, \\ \dfrac{1}{\frac{1}{w(\bar{x})}} \displaystyle\sum_{i=1}^{n+1} f(x_i) \frac{\beta_i}{\bar{x} - x_i} & \text{if } \bar{x} \notin \{x_1, x_2, \cdots, x_{n+1}\}. \end{cases}$$

2. Lagrange basis has a good property, what is called the partition of unity: That is the equality

   $$L_1(x) + L_2(x) + \cdots + L_{n+1}(x) = 1 \quad \text{for all } x \in [a, b].$$

   Its justification is as follows: Suppose we interpolate the constant function $\theta(x) \equiv 1$ expanded in Lagrange basis. Then that must be again $\theta(x)$!

3. Also, because in every expression $L_i(x) = \beta_i \frac{w(x)}{x-x_i}$ we have $w(x) = (x-x_1)(x-x_2)\cdots(x-x_{n+1})$ as the common factor,

$$\text{if } x \notin \{x_1, x_2, \cdots, x_{n+1}\} \quad 1 = w(x)\left(\frac{\beta_1}{x-x_1} + \frac{\beta_2}{x-x_2} + \cdots + \frac{\beta_{n+1}}{x-x_{n+1}}\right)$$

$$\text{Hence} \quad \text{if } x \notin \{x_1, x_2, \cdots, x_{n+1}\} \quad \frac{1}{w(x)} = \frac{\beta_1}{x-x_1} + \frac{\beta_2}{x-x_2} + \cdots + \frac{\beta_{n+1}}{x-x_{n+1}}.$$

In fact, this was to establish the partial fraction

$$\text{if } x \notin \{x_1, x_2, \cdots, x_{n+1}\} \quad \frac{\beta_1}{x-x_1} + \frac{\beta_2}{x-x_2} + \cdots + \frac{\beta_{n+1}}{x-x_{n+1}} = \frac{1}{(x-x_1)(x-x_2)\cdots(x-x_{n+1})}.$$

In any case, if $\bar{x} \notin \{x_1, x_2, \cdots, x_{n+1}\}$ we let

$$t_i = t_i(\bar{x}) = \frac{\beta_i}{\bar{x}-x_i} \in \mathbb{R} \quad \text{then} \quad \frac{1}{w(\bar{x})} = \sum_{k=1}^{n+1} t_k.$$

4. In conclusion,

$$q(\bar{x}) = \begin{cases} f(x_i) & \text{if } \bar{x} \in \{x_1, x_2, \cdots, x_{n+1}\} \text{ and } \bar{x} = x_i, \\ \dfrac{1}{\sum_{k=1}^{n+1} t_k} \displaystyle\sum_{i=1}^{n+1} f(x_i) t_i & \text{if } \bar{x} \notin \{x_1, x_2, \cdots, x_{n+1}\} \end{cases}$$

After all, we have formulated the expression as this weighted sum, though the weight t_i may be negative.

Although weights have signs, we do not worry about the loss of digits in sum of weights, or we do not worry if $\sum_{k=1}^{n+1} \hat{t}_k \simeq 0$, since it is the value $\frac{1}{(\bar{x}-x_1)(\bar{x}-x_2)\cdots(\bar{x}-x_{n+1})}$.

- We can use any normalization

$$\tilde{t}_i = \frac{t_i}{T} \quad \text{for any } T \neq 0.$$

- Noticing that t_i can be of very large scale if x is close to x_i, we can safely normalize in the way

$$\hat{t}_i = \frac{\frac{\beta_i}{B}}{\frac{x-x_i}{M}},$$

where we may choose

$$B = \max_i\{|\beta_i|\}, \quad M = \left(\min_i\{|x-x_i|\}\right)^{1/2}$$

There will not be the overflow. Finally

$$t_i^* = \frac{\hat{t}_i}{\sum_{k=1}^{n+1} \hat{t}_k}$$

- Then we compute

$$q(\bar{x}) = \begin{cases} f(x_i) & \text{if } \bar{x} \in \{x_1, x_2, \cdots, x_{n+1}\} \text{ and } \bar{x} = x_i, \\ \displaystyle\sum_{i=1}^{n+1} f(x_i) t_i^* & \text{if } \bar{x} \notin \{x_1, x_2, \cdots, x_{n+1}\} \end{cases}$$

(blank)

Chapter 6

The remainder $R(x)$

1. As a matter of fact (by Stone-Weirstrass Theorem), for any continuous function f on the compact interval $[a, b]$,

   there exists $p_n \in \mathcal{P}^{<n+1}([a, b])$ such that p_n uniformly converges to f as $n \to \infty$.

2. Because, our procedure of (sampling $\to$ interpolation) for the purpose of <u>approximation</u> is easy and straightforward, we wish the polynomial sequence generated by this method can possibly take the role in the Stone-Weirstrass Theorem.

3. We examine this question of convergence as n increases. The answer to the question is, it is not always so, and thus we explore:

   (a) When it can go wrong and who is our enemy.

   (b) Thumb rules for preventing this to happen.

Our enemy $w(x) = (x - x_1)(x - x_2) \cdots (x - x_{n+1})$

We have seen the polynomial $w(x) = (x - x_1)(x - x_2) \cdots (x - x_{n+1}) \in \mathcal{P}^{<n+2}([a, b])$ of one higher order and this is our enemy.

Suppose f is given and using sampling data the interpolating polynomial $p_{<n+1}$ approximates f sufficiently well.

I can always make a worse approximation problem by considering another function g:

$$g(x) = f(x) + Aw(x) \quad \text{for any } A \in \mathbb{R}$$

Because the sampling data are still same, i.e.,

$$g(x_i) = f(x_i) \quad \text{for every } i$$

the interpolating polynomial $p_{<n+1}$ will remain same for g, and the approximation is as bad as $Aw(x)$ amounts.

The very first problem

The very first problem we must be able to control the remainder is

$$\text{when the function } f = w \text{ and sampling data are } (x_1, 0), (x_2, 0), \cdots, (x_{n+1}, 0)$$

In this case, the interpolating polynomial is the zero function in $\mathcal{P}^{<n+1}$, and the remainder

$$R(x) = w(x) - p_{<n+1}(x)$$

is the w itself.

Of course, this remainder is not conflict with the Taylor's theorem or 1d remainder theorem.

Indeed, by the 1d remainder theorem

$$R(x) = (x - x_1)(x - x_2)(x - x_3) \cdots (x - x_{n+1})\big([x_1, x_2, \cdots, x_{n+1}, x], w\big)$$

and check the definition

$$\big([x_1, x_2, \cdots, x_{n+1}, x], w\big) = \frac{1}{\sqrt{n+2}} \int_{\Lambda^{n+1} \subset \mathbb{R}^{n+2}} \frac{d^{n+1}w}{dx^{n+1}}\big(\lambda_1 x_1 + \cdots \lambda_{n+1} x_{n+1} + \lambda_{n+2} x\big) dS(\lambda)$$

where

$$\frac{d^{n+1}w}{dx^{n+1}}(x) = \frac{d^{n+1}}{dx^n}(x - x_1)(x - x_2)(x - x_3) \cdots (x - x_{n+1}) = (n+1)! \quad (const.)$$

and the volume of the regular $n+1$ dimensional simplex

$$\frac{1}{\sqrt{n+2}} \int_{\Lambda^{n+1} \subset \mathbb{R}^{n+2}} dS(\lambda) = \frac{1}{(n+1)!},$$

so $\big([x_1, x_2, \cdots, x_{n+1}, x], w\big) \equiv 1$. In other words, the remainder $R(x) = w(x)$.

To monitor how the remainder amounts for this case,

The constant $M := \sup\limits_{x \in [a,b]} |w(x)|$

1. We define $M := \sup\limits_{x \in [a,b]} |w(x)|$.

2. M is dependent on the $[a, b]$, and the choice $(x_i)_{i=1}^{n+1}$ contained in $[a, b]$.

Polynomial is bad if the interval length exceeds 2.

The correct statement is that polynomial is bad if the interval length exceeds 2, under the assumption that the all of its roots are in the interval.

We consider the interval $[0, 2a]$ for some $a > 0$ and the simple equi-distance sampling case, where sampling points are

$$0, h, 2h, 3h, \cdots nh = 2a$$

Calculation of $\left|w\left(\frac{h}{2}\right)\right|$: equi-distance sampling

Instead of computing the supremum M, we go for easier calculations of $K = \left|w\left(\frac{h}{2}\right)\right|$. Note the equi-distance $h = \frac{2a}{n}$.

$$\begin{aligned} K &= \frac{a}{n} \times \frac{a}{n} \times \frac{3a}{n} \times \cdots \times \frac{(2n-1)a}{n} \\ &= \frac{a^{n+1}}{n^{n+1}}\, 1 \times 3 \times \cdots \times (2n-1). \end{aligned}$$

Because

$$\begin{aligned} 1 \times 2 \times 4 \times \cdots \times 2n-2 \quad &< \quad 1 \times 3 \times \cdots \times (2n-1) \quad < \quad 2 \times 4 \times \cdots \times 2n, \\ (2n-1)! \quad &< \quad \Big(1 \times 3 \times \cdots \times (2n-1)\Big)^2 \quad < \quad (2n)! \end{aligned}$$

Thus,

$$\frac{a^{n+1}}{n^{n+1}}\sqrt{(2n-1)!} \quad < \quad K \quad < \quad \frac{a^{n+1}}{n^{n+1}}\sqrt{(2n)!}.$$

It is known that for $n \geq 1$

$$\sqrt{2\pi}\sqrt{2n}\Big(\frac{2n}{e}\Big)^{2n} \quad < \quad (2n)! \quad < \quad e\sqrt{2\pi}\sqrt{2n}\Big(\frac{2n}{e}\Big)^{2n}.$$

This gives that

$$\frac{a^{n+1}}{n^{n+1}}\frac{1}{\sqrt{2n}} \times \sqrt{\sqrt{2\pi}\sqrt{2n}\Big(\frac{2n}{e}\Big)^{2n}} \quad < \quad K \quad < \quad \frac{a^{n+1}}{n^{n+1}} \times \sqrt{e\sqrt{2\pi}\sqrt{2n}\Big(\frac{2n}{e}\Big)^{2n}}$$

This simplifies to that

$$\Big(\frac{2a}{e}\Big)^{n+1} \times n^{-5/4} \times \frac{e}{2\sqrt{2}}(4\pi)^{\frac{1}{4}} \quad < \quad K \quad < \quad \Big(\frac{2a}{e}\Big)^{n+1} \times n^{-3/4} \times \frac{e^{3/2}}{2}(4\pi)^{\frac{1}{4}}.$$

Since $e = 2.718...$ the interval length $2a$ should not exceed 2.718...

This tells us that

1. Under the assumption that all roots should be contained in the interval,
2. Polynomial of high order exhibits many reflection points and oscillations in the interval and the value become enormous near end points if length is long enough.
3. If there is 100 sampling data in the interval of length 10, it may not be a good idea to consider 99-th order polynomial in that interval, but
4. We divide it into 10 pieces, and in each of pieces we take lower order polynomials and assemble them all together.

Next, for given length, we examine the dependency of M on the choice of sampling points $(x_i)_{i=1}^{n+1}$ further.

More specifically, we consider the special case where we can choose where to sample, and if this is the case we answer to that what could be the optimal choice.

Polynomial that is like Trigonometries: Chebyshev polynomial

We give a minimal introduction to the Chebyshev polynomial.

Let $I = [-a, a]$, $\tilde{I} = [0, \pi]$. We consider the parametrization by a bijection from $[-a, a]$:

$$\theta : [-a, a] \to [0, \pi], \quad \theta(x) = \arccos\left(\frac{x}{a}\right).$$

Definition 1. *For each $k = 0, 1, 2, \cdots$*

$$T_k(x) := \cos\big(k\theta(x)\big).$$

Proposition 2. *For each $k = 0, 1, 2, \cdots$,*

1. *T_k has k zeroes in $[-a, a]$.*
2. *For $k \geq 1$, T_k attains -1, the minimum, and $+1$, the maximum, alternatingly so that $|T_k(x)|$ hits 1 exactly $k+1$ times in $[-a, a]$.*
3. *T_k is a k-th order polynomial.*
4. *For $k \geq 1$, the coefficient of the k-th (highest) order term is $\frac{1}{2}\left(\frac{2}{a}\right)^k$.*

Proof. 1. In case $k = 0$, $T_0(x) \equiv 1$ and it has no zero. For $k \geq 1$, T_k attains zero only at x such that

$$k\theta(x) = \frac{\pi}{2} + m\pi$$

for m integer. We observe that those integers m so that $\phi_m = \frac{1}{k}\left(\frac{\pi}{2} + m\pi\right) \in [0, \pi]$, i.e. those integers so that $r_m = \theta^{-1}(\phi_m) \in [-a, a]$, are $m = 0, 1, 2, \cdots, k-1$. Note (ϕ_m) is strictly increasing in m and $\theta(x)$ is a bijection. Thus, $r_0, r_1, \cdots, r_{k-1}$ are distinct k points in $[-a, a]$.

2. $T_k(x) \in [-1, 1]$, the range of the cosine function. For $k \geq 1$, $|T_k(x)|$ hits 1 only at x such that
$$k\theta(x) = m\pi$$
for m integer. We observe that those integers m so that $\tilde{\phi}_m = \frac{m\pi}{k} \in [0, \pi]$, i.e. those integers so that $t_m = \theta^{-1}(\tilde{\phi}_m) \in [-a, a]$, are $m = 0, 1, 2, \cdots, k$. Note $(\tilde{\phi}_m)$ is strictly increasing in m and $\theta(x)$ is a decreasing bijection. Thus, $t_0, t_1, \cdots, t_k$ are distinct $k+1$ points such that
$$t_0 > t_1 > \cdots > t_k$$
and $T_k(t_m) = \cos(m\pi) = (-1)^m$.

3. For $k = 0$, $T_0(x) \equiv 1$ is 0-th order polynomial. For $k = 1$, $T_1(x) = \frac{x}{a}$. Now, we use the trigonometric identity for $k \geq 1$. For notational convenience write $\theta = \theta(x)$, then
$$\begin{aligned} T_{k+1}(x) &= \cos\Big(k\theta + \theta\Big) = \cos k\theta \cos\theta - \sin k\theta \sin\theta \\ T_{k-1}(x) &= \cos\Big(k\theta - \theta\Big) = \cos k\theta \cos\theta + \sin k\theta \sin\theta \\ \implies \quad & T_{k+1}(x) + T_{k-1}(x) = 2T_k(x)\cos(\theta(x)) = 2T_k(x)\frac{x}{a} \\ \implies \quad & T_{k+1}(x) = 2T_k(x)\frac{x}{a} - T_{k-1}(x). \end{aligned}$$
Therefore, if T_0, T_1, $\cdots$, T_{k-1} are polynomials of order $0, 1, \cdots, k-1$ respectively, then $T_k(x)$ is a k-th order polynomial.

4. The coefficient of the highest order term of $T_1(x)$ is $\frac{1}{a}$, and from the last recursive formula, that of $T_k(x)$ for $k \geq 1$ is
$$\frac{1}{a}\Big(\frac{2}{a}\Big)^{k-1} = \frac{1}{2}\Big(\frac{2}{a}\Big)^{k}.$$

□

Calculation of M: sampling at chebyshev zeroes

Now, we can compute $M := \sup\limits_{x \in [-a,a]} \big|(x - r_0)(x - r_1)\cdots(x - r_n)\big|$. Since r_i are $(n+1)$ distinctive zeroes of $(n+1)$-th order polynomials T_{n+1} and w, and the coefficient of the highest order of $w(x)$ is 1, $w(x)$ must be
$$w(x) = 2\Big(\frac{a}{2}\Big)^{n+1} T_{n+1}(x).$$
Since the maximum of $|T_{n+1}(x)| = 1$, we conclude that $M = 2\Big(\frac{a}{2}\Big)^{n+1}$.

Corollary 3. *The constant* $M = 2\Big(\frac{a}{2}\Big)^{n+1}$.

Remark 6.1. Therefore, if the interval length is 2, or $a = 1$, $M = \sup_{x \in [-a,a]} |w(x)|$ will geometrically tends to 0 as n increases.

Remark 6.2. In fact, one can also prove that, there exists no polynomial of the form
$$(x - x_1)(x - x_2)\cdots(x - x_{n+1})$$
whose constant M is strictly less than $2\Big(\frac{a}{2}\Big)^{n+1}$.

In the standard library for scientific computation in our computer, unless there is a specific better algorithm, most of known functions are implemented as piecewise polynomial based on Chebyshev polynomials, and guarantee the double precision of relative error scale

$$10^{-16} \quad \Big(\simeq 2^{-53} \simeq \eta_{machine} \Big)$$

Runge's example and Hilbert space techniques

Suppose $a = 1$. The interval lenth $2a = 2$. For this interval:

1. One can prove in the chebyshev zero case that the interpolating polynomial p_n at chebyshev zeroes does converge uniformly to a given function f, if f has a certain differentiability, say $f \in C^2([-1,1])$. This proof needs Hilbert space techniques.

2. Runge's example and conceivable danger we can see.

   By contrast, in the equi-distance case, Runge found an example of infinitely smooth function on $[-1,1]$, where the contribution from $\big|f[x_1, x_2, \cdots, x_{n+1}, x]\big|$ increases so fast in n that it overwhelms the geometric decrement of the contribution from $|w(x)|$.

$$\begin{aligned}
f(x) &= \frac{1}{1+(5x)^2} = \frac{1}{2i}\Big(\frac{1}{5x-i} - \frac{1}{5x+i}\Big) = \frac{1}{5\times 2i}\Big(\frac{1}{x-i/5} - \frac{1}{x+i/5}\Big) \\
\frac{d^{n+1}f}{dx^{n+1}}(x) &= \frac{1}{5\times 2i}(-1)^{n+1}(n+1)!\Big(\frac{1}{(x-i/5)^{n+2}} - \frac{1}{(x+i/5)^{n+2}}\Big) \\
&= \frac{1}{5\times 2i}(-1)^{n+1}(n+1)!5^{n+2}\Big(\frac{1}{(5x-i)^{n+2}} - \frac{1}{(5x+i)^{n+2}}\Big) \\
&= \frac{(-1)^{n+1}(n+1)!5^{n+1}}{(1+(5x)^2)^{n+2}}\frac{1}{2i}\big(z^{n+2} - \bar{z}^{n+2}\big), \quad z = 5x+i, \quad x \in \mathbb{R} \\
&= \frac{(-1)^{n+1}(n+1)!5^{n+1}}{(1+(5x)^2)^{n+2}}\rho(x)^{n+2}\frac{1}{2i}\big(e^{i(n+2)\theta(x)} - e^{-i(n+2)\theta(x)}\big), \quad z = 5x+i = \rho(x)e^{i\varphi(x)} \\
&= \frac{(-1)^{n+1}(n+1)!5^{n+1}}{\big(\sqrt{1+(5x)^2}\big)^{n+2}}\sin\big((n+2)\varphi(x)\big).
\end{aligned}$$

   This derivative formula alone cannot conclude that $\sup\limits_{x\in[-1,1]} |R(x)|$ diverge as n increases, but the factor 5^{n+1} is a conceivable danger: it overwhems the factor $\big(\frac{2}{e}\big)^{n+1}$, i.e., $5 > e$.

Further reading: Atkinson, Introduction to Numerical Analysis, Chapter 4.

At this stage, we wrap it up here, and move on to the next topic.

Chapter 7

Proof of properties of $\big([x_1, x_2, \cdots, x_{\ell+1}], f\big)$

We recall the definition:

Definition 1. *For $f \in C^{n+1}([a,b])$, $0 \le \ell \le n+1$ integer, $x_1, x_2, \cdots, x_{\ell+1} \in [a,b]$,*

if $\ell = 0$ $\qquad \big([x_1], f\big) = f(x_1)$,

if $\ell \ge 1$ $\quad \big([x_1, x_2, \cdots, x_{\ell+1}], f\big) := \dfrac{1}{\sqrt{\ell+1}} \displaystyle\int_{\Lambda^\ell \subset \mathbb{R}^{\ell+1}} \frac{d^\ell f}{dx^\ell}\big(\lambda_1 x_1 + \cdots \lambda_{\ell+1} x_{\ell+1}\big)\, dS(\lambda)$.

The set Λ^ℓ is the regular ℓ-simplex embedded in $\mathbb{R}^{\ell+1}$:

$$= \Big\{(\lambda_1, \lambda_2, \cdots, \lambda_{\ell+1}) \in \mathbb{R}^{\ell+1} \mid \forall e \;\; \lambda_e \ge 0, \quad \sum_{e=1}^{\ell+1} \lambda_e = 1\Big\} \subset \mathbb{R}^{\ell+1}$$

The integral is called the ℓ-th order Newton's divided difference, and denoted by $\big([x_1, x_2, \cdots, x_{\ell+1}], f\big)$, or $[x_1, \cdots, x_{\ell+1}]f$, or $f[x_1, x_2, \cdots, x_{\ell+1}]$.

We again emphasize that, the Newton's divided difference up to ℓ-th order make senses under the assumptions $\ell \le n+1$ and $f \in C^{n+1}([a,b])$ and otherwise, the definition and properties may not be justifiable.

Let n and ℓ are as in the definition. We are left to prove the following four:

1. The value is independent of orders in the choice $[x_1, x_2, \cdots, x_{\ell+1}]$.
2. The key equality holds:
$$f[x_1, x_2, \cdots, x_{\ell+1}]\,(x_{\ell+1} - x_1) = f[x_2, x_3, \cdots, x_{\ell+1}] - f[x_1, x_2, \cdots, x_\ell]. \quad \text{(KEY)}$$
3. Mean Value type evaluation holds:
$$\text{There exists } \xi \in [a,b] \text{ such that} \quad f[x_1, x_2, \cdots, x_{\ell+1}] = \frac{f^{(\ell)}(\xi)}{\ell!}$$
4. The special case $x_1 = x_2 = \cdots = x_{n+1}$ of 1d remainder theorem is precisely Taylor's theorem.

Order independence

Proposition 1. *Let $f \in C^{n+1}([a,b])$, $0 \le \ell \le n+1$ integer, and $x_1, x_2, \cdots, x_{\ell+1} \in [a,b]$. Then, for any permuation $P : \{1, 2, \cdots, \ell+1\} \to \{1, 2, \cdots, \ell+1\}$*

$$f[x_{P(1)}, x_{P(2)}, \cdots, x_{P(\ell+1)}] = f[x_1, x_2, \cdots, x_{\ell+1}].$$

Proof. This is because of nothing but Λ^ℓ is a regular ℓ-simplex. Let θ be any function

$$\theta : \Lambda^\ell \to \mathbb{R}$$

and we verify that the integral of any θ over the regular ℓ-simplex

$$\int_{\Lambda^\ell} \theta(\lambda)\, dS(\lambda)$$

is independent of the permutation of the variable.

Let $(t_1, \cdots, t_{\ell+1}) \in \Lambda^\ell$ and parametrization

$$\phi : (t_1, t_2, \cdots, t_{\ell+1}) \mapsto (\lambda_1, \lambda_2, \cdots, \lambda_{\ell+1}) = (t_{P(1)}, t_{P(2)}, \cdots, t_{P(\ell+1)}).$$

Then, by change of variable formula,

$$\int_{\Lambda^\ell} \theta(\lambda_1, \lambda_2, \cdots, \lambda_{\ell+1})\, dS(\lambda) = \int_{\phi^{-1}(\Lambda^\ell)} \theta \circ \phi(t_1, t_2, \cdots, t_{\ell+1}) \left|\det D\phi\right| dS(t).$$

We see that

1. $\phi^{-1}(\Lambda^\ell) = \Lambda^\ell$, and
2. $\left|\det D\phi\right| = \left|\det P\right| = 1$, where P is the permutation matrix.

Hence,

$$\int_{\Lambda^\ell} \theta(\lambda_1, \lambda_2, \cdots, \lambda_{\ell+1})\, dS(\lambda) = \int_{\Lambda^\ell} \theta(t_{P(1)}, t_{P(2)}, \cdots, t_{P(\ell+1)})\, dS(t).$$

The assertion follows by renaming the variable in the left-hand-side,

$$\int_{\Lambda^\ell} \theta(t_1, t_2, \cdots, t_{\ell+1})\, dS(t) = \int_{\Lambda^\ell} \theta(t_{P(1)}, t_{P(2)}, \cdots, t_{P(\ell+1)})\, dS(t).$$

Having verified this, it is straightforward to establish that

$$\begin{aligned}
f[x_{P(1)}, x_{P(2)}, \cdots, x_{P(\ell+1)}] &= \frac{1}{\sqrt{\ell+1}} \int_{\Lambda^\ell} \frac{d^\ell f}{dx^\ell}\big(\lambda_1 x_{P(1)} + \cdots + \lambda_{\ell+1} x_{P(\ell+1)}\big)\, dS(\lambda) \\
&= \frac{1}{\sqrt{\ell+1}} \int_{\Lambda^\ell} \frac{d^\ell f}{dx^\ell}\big(\lambda_{P(1)} x_{P(1)} + \cdots + \lambda_{P(\ell+1)} x_{P(\ell+1)}\big)\, dS(\lambda) \\
&= \frac{1}{\sqrt{\ell+1}} \int_{\Lambda^\ell} \frac{d^\ell f}{dx^\ell}\big(\lambda_1 x_1 + \cdots + \lambda_{\ell+1} x_{\ell+1}\big)\, dS(\lambda) \\
&= f[x_1, x_2, \cdots, x_{\ell+1}].
\end{aligned}$$

□

Key equality

Proposition 2. *Let $f \in C^{n+1}([a,b])$, $1 \le \ell \le n+1$ integer, and $x_1, x_2, \cdots, x_{\ell+1} \in [a,b]$. Then,*

$$f[x_1, x_2, \cdots, x_{\ell+1}]\ (x_{\ell+1} - x_1) = f[x_2, x_3, \cdots, x_{\ell+1}] - f[x_1, x_2, \cdots, x_\ell]. \qquad \text{(KEY)}$$

Before we prove the proposition, we conduct following calculations:

1. Change of Variable.

   For $\ell \ge 1$, we define the set $\mathcal{T}^\ell = \big\{(\tau_2, \tau_3, \cdots, \tau_{\ell+1}) \in \mathbb{R}^\ell \mid 0 \le \tau_2 + \cdots + \tau_{\ell+1} \le 1\big\}$

   and we let the parametrization from $\mathcal{T}^\ell$ to Λ^ℓ

   $\varphi : (\tau_2, \tau_3, \cdots, \tau_{\ell+1}) \quad \mapsto \quad \big(\lambda_1, \lambda_2, \cdots, \lambda_{\ell+1}\big) = \big((1-\tau_2-\tau_3-\cdots-\tau_{\ell+1}), \tau_2, \tau_3, \cdots, \tau_{\ell+1}\big).$

   The differential $D\varphi$ is then $(\ell+1) \times \ell$ matrix

$$\begin{pmatrix} -1 & -1 & -1 & \cdots & -1 \\ 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{pmatrix}$$

   Then, by change of variable formula, for $\ell \ge 1$,

$$\begin{aligned} f[x_1, x_2, \cdots, x_{\ell+1}] &= \frac{1}{\sqrt{\ell+1}} \int_{\Lambda^\ell \subset \mathbb{R}^{\ell+1}} \frac{d^\ell f}{dx^\ell}\big(\lambda_1 x_1 + \cdots + \lambda_{\ell+1} x_{\ell+1}\big)\, dS(\lambda) \\ &= \frac{1}{\sqrt{\ell+1}} \int_{\varphi^{-1}(\Lambda^\ell)} \frac{d^\ell f}{dx^\ell}\big((1-\tau_2-\tau_3-\cdots-\tau_{\ell+1})x_1 + \tau_2 x_2 + \cdots + \tau_{\ell+1} x_{\ell+1}\big) \sqrt{\big|\det D\varphi^T D\varphi\big|}\, d\tau \\ &= \int_{\mathcal{T}^\ell} \frac{d^\ell f}{dx^\ell}\big((1-\tau_2-\tau_3-\cdots-\tau_{\ell+1})x_1 + \tau_2 x_2 + \cdots + \tau_{\ell+1} x_{\ell+1}\big)\, d\tau \end{aligned}$$

2. Fubini

   The integral in $\mathcal{T}^\ell \subset \mathbb{R}^\ell$ is conducted by iterated 1-dimensional integrations,

$$\begin{aligned} f[x_1, x_2, \cdots, x_{\ell+1}] &= \int_{\mathcal{T}^\ell} \frac{d^\ell f}{dx^\ell}\big((1-\tau_2-\tau_3-\cdots-\tau_{\ell+1})x_1 + \tau_2 x_2 + \cdots + \tau_{\ell+1} x_{\ell+1}\big)\, d\tau \\ &= \int_{\tau_2=0}^{1} \int_{\tau_3=0}^{1-\tau_2} \int_{\tau_4=0}^{1-\tau_2-\tau_3} \cdots \int_{\tau_{\ell+1}=0}^{1-\tau_2-\cdots-\tau_\ell} \\ &\qquad \frac{d^\ell f}{dx^\ell}\big((1-\tau_2-\tau_3-\cdots-\tau_{\ell+1})x_1 + \tau_2 x_2 + \cdots + \tau_{\ell+1} x_{\ell+1}\big) \\ &\quad d\tau_{\ell+1} d\tau_\ell \cdots d\tau_2. \end{aligned}$$

3. The ℓ-volume of the Λ^ℓ.

   We verify by the iterated integral

$$\int_{\tau_2=0}^{1}\int_{\tau_3=0}^{1-\tau_2}\int_{\tau_4=0}^{1-\tau_2-\tau_3}\cdots\int_{\tau_{\ell+1}=0}^{1-\tau_2-\cdots-\tau_\ell} 1\, d\tau_{\ell+1}d\tau_\ell\cdots d\tau_2 = \frac{1}{\ell!}$$

Now, we are ready for the proof of Proposition 2.

proof of Proposition 2. Thanks to the order independence,

$$\begin{aligned}
&f[x_2, x_3, x_4, \cdots, x_{\ell+1}] - f[x_1, x_2, x_3, \cdots, x_\ell]\\
=&f[x_{\ell+1}, x_2, x_3, \cdots, x_\ell] - f[x_1, x_2, x_3, \cdots, x_\ell]\\
=&\int_{\tau_2=0}^{1}\int_{\tau_3=0}^{1-\tau_2}\int_{\tau_4=0}^{1-\tau_2-\tau_3}\cdots\int_{\tau_\ell=0}^{1-\tau_2-\cdots-\tau_{\ell-1}}\\
&\qquad\frac{d^{\ell-1}f}{dx^{\ell-1}}\big((1-\tau_2-\tau_3-\cdots-\tau_\ell)x_{\ell+1}+\tau_2x_2+\tau_3x_3+\cdots+\tau_\ell x_\ell\big)\\
&\qquad-\frac{d^{\ell-1}f}{dx^{\ell-1}}\big((1-\tau_2-\tau_3-\cdots-\tau_\ell)x_1+\tau_2x_2+\tau_3x_3+\cdots+\tau_\ell x_\ell\big)\\
&d\tau_\ell\cdots d\tau_2\\
=&\int\int\cdots\int A(\tau_2, \tau_3, \cdots, \tau_\ell)\, d\tau_\ell\cdots d\tau_2
\end{aligned}$$

Note that the integrand A can be written as

$$\left[\frac{d^{\ell-1}f}{dx^{\ell-1}}\big((1-\tau_2-\tau_3-\cdots-\tau_\ell-\tau_{\ell+1})x_1+\tau_2x_2+\tau_3x_3+\cdots+\tau_\ell x_\ell+\tau_{\ell+1}x_{\ell+1}\big)\right]_{\tau_{\ell+1}=0}^{1-\tau_2-\tau_3-\cdots-\tau_\ell}$$

Checking that for each fixed $\tau_2, \tau_3, \cdots, \tau_\ell$, the map

$$\tau_{\ell+1}\mapsto\frac{d^{\ell-1}f}{dx^{\ell-1}}\big((1-\tau_2-\tau_3-\cdots-\tau_\ell-\tau_{\ell+1})x_1+\tau_2x_2+\tau_3x_3+\cdots+\tau_\ell x_\ell+\tau_{\ell+1}x_{\ell+1}\big)$$

is a continously differentiable function, Fundamental Theorem of Calculus implies that the integrand A can be written as

$$\begin{aligned}
&A(\tau_2, \tau_2, \cdots, \tau_\ell) = \int_{\tau_{\ell+1}=0}^{1-\tau_2-\tau_2-\cdots-\tau_\ell}\\
&\frac{d}{d\tau_{\ell+1}}\frac{d^{\ell-1}f}{dx^{\ell-1}}\big((1-\tau_2-\tau_3-\cdots-\tau_\ell-\tau_{\ell+1})x_1+\tau_2x_2+\tau_3x_3+\cdots+\tau_\ell x_\ell+\tau_{\ell+1}x_{\ell+1}\big)\, d\tau_{\ell+1}.
\end{aligned}$$

By chain rule,

$$\begin{aligned}
&A(\tau_2, \tau_2, \cdots, \tau_\ell) = \int_{\tau_{\ell+1}=0}^{1-\tau_2-\tau_3-\cdots-\tau_\ell}\\
&(x_{\ell+1}-x_1)\times\frac{d}{dx}\frac{d^{\ell-1}f}{dx^{\ell-1}}\big((1-\tau_2-\tau_3-\cdots-\tau_\ell-\tau_{\ell+1})x_1+\tau_2x_2+\tau_3x_3+\cdots+\tau_\ell x_\ell+\tau_{\ell+1}x_{\ell+1}\big)\, d\tau_{\ell+1}.
\end{aligned}$$

We have proven that

$$
\begin{aligned}
&\int\int\cdots\int A(\tau_2,\tau_3,\cdots,\tau_\ell)\,d\tau_\ell\cdots d\tau_2\\
=&(x_{\ell+1}-x_1)\times\int_{\tau_2=0}^{1}\int_{\tau_3=0}^{1-\tau_2}\int_{\tau_4=0}^{1-\tau_2-\tau_3}\cdots\int_{\tau_\ell=0}^{1-\tau_2-\cdots-\tau_{\ell-1}}\int_{\tau_{\ell+1}=0}^{1-\tau_2-\tau_3-\cdots-\tau_\ell}\\
&\frac{d^\ell f}{dx^\ell}\big((1-\tau_2-\tau_3-\cdots-\tau_\ell-\tau_{\ell+1})x_1+\tau_2x_2+\tau_3x_3+\cdots+\tau_\ell x_\ell+\tau_{\ell+1}x_{\ell+1}\big)\\
&d\tau_{\ell+1}d\tau_\ell\cdots d\tau_2
\end{aligned}
$$

which is precisely the (LHS)

$$f[x_1,x_2,\cdots,x_{\ell+1}]\ (x_{\ell+1}-x_1).$$

□

Mean Value type evaluation

Proposition 3. *Let $f \in C^{n+1}([a,b])$, $0 \le \ell \le n+1$ integer, and $x_1, x_2, \cdots, x_{\ell+1} \in [a,b]$. Then,*

$$\text{there exists } \xi \in [a,b] \text{ such that } \quad f[x_1, x_2, \cdots, x_{\ell+1}] = \frac{f^{(\ell)}(\xi)}{\ell!}$$

Proof. For $\ell = 0$, the conclusion is trivially true. We assume $\ell \ge 1$ from now on.

1. We first verify that the map

$$A : (\lambda_1, \lambda_2, \cdots, \lambda_{\ell+1}) \mapsto \frac{1}{\sqrt{\ell+1}} \frac{d^\ell f}{dx^\ell}\big(\lambda_1 x_1 + \cdots \lambda_{\ell+1} x_{\ell+1}\big)$$

   is continuous on the compact set Λ^ℓ. Therefore, there exists $\lambda_*, \lambda^* \in \Lambda^\ell$ where the map attains the minimum and maximum. Let also A_*, A^* denote the minimum and maximum value respectively.

2. Let us denote $x(\lambda) := \lambda_1 x_1 + \cdots \lambda_{\ell+1} x_{\ell+1}$. The convex combination $x(\lambda)$ of $x_1, x_2, \cdots, x_{\ell+1} \in [a,b]$ must also be in $[a,b]$.

3. Now,

$$\int_{\Lambda^\ell} A_* \, dS(\lambda) \le \int_{\Lambda^\ell} \frac{1}{\sqrt{\ell+1}} \frac{d^\ell f}{dx^\ell}\big(x(\lambda)\big) \, dS(\lambda) \le \int_{\Lambda^\ell} A^* \, dS(\lambda).$$

   In other words,

$$\frac{A_* \sqrt{\ell+1}}{\ell!} \le f[x_1, x_2, \cdots, x_{\ell+1}] \le \frac{A^* \sqrt{\ell+1}}{\ell!}.$$

4. In the convex set Λ^ℓ, the line segment J joining λ_* and λ^* is contained. In the special case $\lambda_* = \lambda^*$, J is a one point set. Then, we note the values of A along the line segment J

$$\alpha : [0,1] \ni t \quad \mapsto \quad \frac{1}{\sqrt{\ell+1}} \frac{d^\ell f}{dx^\ell}\big((1-t)\lambda_* + t\lambda^*\big)$$

   must attain all values in $[A_*, A^*]$ by Intermediate Value Theorem.

5. The value $\frac{\ell!}{\sqrt{\ell+1}} f[x_1, x_2, \cdots, x_{\ell+1}] \in [A_*, A^*]$ must have been attained as $\alpha(\hat{t})$ at some

$$\hat{t} \in [0,1], \quad \xi = x\big((1-\hat{t})\lambda_* + \hat{t}\lambda^*\big) \in [a,b], \quad \text{i.e.}$$

$$\frac{\ell!}{\sqrt{\ell+1}} f[x_1, x_2, \cdots, x_{\ell+1}] = \alpha(\hat{t}) = \frac{1}{\sqrt{\ell+1}} \frac{d^\ell f}{dx^\ell}(\xi).$$

   The conclusion then follows.

□

Reduction to the Taylor's Theorem

Proposition 4. *Let $f \in C^{n+1}([a,b])$, and $x_0 \in [a,b]$. Then for $x \in [a,b]$,*

(x_1 is n+1 times repeated) $$f[x_1, x_1, \cdots, x_1, x] = \int_{\tau=0}^{1} \frac{(1-\tau)^n}{n!} \frac{d^{n+1}f}{dx^{n+1}}\Big((1-\tau)x_1 + \tau x\Big) d\tau$$

Proof. The reduction is justified by following calculations: If $x_1 = x_2 = \cdots = x_{n+1}$,

$$\begin{aligned}
&R(x) = (x-x_1)(x-x_2)\cdots(x-x_{n+1}) f[x_1, x_2, \cdots, x_{n+1}, x] \\
=&(x-x_1)^{n+1} \times \int_{\mathcal{T}^{n+1}} \frac{d^{n+1}f}{dx^{n+1}}\big((1-\tau_2-\tau_3-\cdots-\tau_{n+2})x_1 + \tau_2 x_2 \cdots \tau_{n+1}x_{n+1} + \tau_{n+2}x\big)\, d\tau \\
=&(x-x_1)^{n+1} \times \int_{\mathcal{T}^{n+1}} \frac{d^{n+1}f}{dx^{n+1}}\big((1-\tau_{n+2})x_1 + \tau_{n+2}x\big)\, d\tau \\
=&(x-x_1)^{n+1} \times \\
&\int_{\tau_{n+2}=0}^{1} \frac{d^{n+1}f}{dx^{n+1}}\big((1-\tau_{n+2})x_1 + \tau_{n+2}x\big) \int_{\tau_{n+1}=0}^{1-\tau_{n+2}} \int_{\tau_n=0}^{1-\tau_{n+2}-\tau_{n+1}} \cdots \int_{\tau_2=0}^{1-\tau_{n+2}-\cdots-\tau_3} 1\, d\tau_2 d\tau_3 \cdots d\tau_{n+2} \\
=&(x-x_1)^{n+1} \times \int_{\tau_{n+2}=0}^{1} \frac{d^{n+1}f}{dx^{n+1}}\big((1-\tau_{n+2})x_1 + \tau_{n+2}x\big) \times \frac{(1-\tau_{n+2})^n}{n!}\, d\tau_{n+2}.
\end{aligned}$$

By renaming the variable $\tau = \tau_{n+2}$,

$$R(x) = (x-x_1)^{n+1} \int_{\tau=0}^{1} \frac{(1-\tau)^n}{n!} \frac{d^{n+1}f}{dx^{n+1}}\Big((1-\tau)x_1 + \tau x\Big)\, d\tau.$$

□

Chapter 8

Algebraic equation solving

We are to solve a system of n equations on n unknowns: for instance

$$\begin{aligned}
\sin(x_1 + x_3^2) + \exp(x_5)\cos(x_2 x_4) &= 3, \\
x_3^2 + x_4^2 &= 1, \\
x_1^4 - x_2^3 - 4x_3 + x_2 &= -7, \\
\sinh(x_3 - x_1) + \cosh(x_2 + x_5) &= 3, \\
x_5 + \log(x_1) &= -2.
\end{aligned}$$

For $\mathbf{f} : E \subset \mathbb{R}^n \to \mathbb{R}^n$ and $\mathbf{y} \in \mathbb{R}^n$ we solve

$$\mathbf{f}(\mathbf{x}) = \mathbf{y}, \quad \mathbf{x} \in \mathbb{R}^n$$

that is a system of equations

$$\begin{aligned}
f_1(x_1, x_2, \cdots, x_n) &= y_1, \\
f_2(x_1, x_2, \cdots, x_n) &= y_2, \\
f_3(x_1, x_2, \cdots, x_n) &= y_3, \\
&\vdots \\
f_n(x_1, x_2, \cdots, x_n) &= y_n.
\end{aligned}$$

We may of course let $\tilde{\mathbf{f}}(\mathbf{x}) = \mathbf{f}(\mathbf{x}) - \mathbf{y}$ and solve $\tilde{\mathbf{f}}(\mathbf{x}) = 0$.

We will say the rootfinding problem for given $\mathbf{y}$ for the former and the zerofinding problem for the latter.

Nonlinear equations

For a given nonlinear system of equations, we do not know in advance in general,

1. whether any solution exists or not,
2. if exists, whether there are many solutions or not.

The two ways to deal with difficult problem.

One way to solve difficult problem is to come up with a great idea.

Another way is, we instead solve a different problem that is not difficult. A reasonable idea is to make this nonlinear problem into an artificial linear problem.

Suppose we want to compute the $\sqrt{5}$, or we want to solve

$$x^2 = 5.$$

Since we can solve $ax = 5$ $(a \neq 0)$, namely $x = \frac{5}{a}$, we artificially pose a problem

$$\bar{x} \times x = 5,$$

where $\bar{x}$ is to be regarded as a *coefficient*. Now, for given $\bar{x}$, define

$$\frac{5}{\bar{x}} =: \eta(\bar{x}).$$

We now hope that we are very lucky that it happen to be $x = \eta(\bar{x}) = \bar{x}$. More rational expectation is to hope the iteration

$$x_{n+1} = \eta(x_n)$$

converges. If so, assuming η is continuous, this implies that at the limit value x_*, $x_* = \eta(x_*)$ and we found the solution.

The art of choosing $\bar{x}$ part and x part in nonlinear equations.

The first objective of this crude procedure can be the following.

1. We choose $\bar{x}$ part and x part in nonlinearities in such a way so that the solution $x = \eta(\bar{x})$ of the artificial solvable problem falls into a bounded controlled region.
2. More specifically, this first objective is to come up with choice of $\bar{x}$, x, and the candidate set $K \subset \mathbb{R}^n$ that is compact and convex such that
$$\eta(K) \subset K.$$
3. our example $\eta(\bar{x}) = \frac{5}{\bar{x}}$?

A better iteration?

We return to the problem of solving $x^2 = 5$. For some reason, instead of $\frac{5}{\bar{x}}$, we set
$$\eta(\bar{x}) = \frac{1}{2}\Big(\bar{x} + \frac{5}{\bar{x}}\Big)$$
and we run the iteration
$$x_{n+1} = \eta(x_n),$$
hoping the generated sequence converges.

There is the canonical linear modification for $f(x) = 0$

1. For a given nonlinear equation, its zerofinding problem is difficult to solve.
2. We happen to know that $\bar{x}$ is <u>a good guess</u> for a zero.
3. Then, near $\bar{x}$, $f(x)$ is close to the linear function
$$f(x) = p_1(x) + R(x) \simeq p_1(x) = f(\bar{x}) + f'(\bar{x})(x - \bar{x}),$$
and here the assumption that $\underline{f \in C^1}$ justifies the approximation.
4. While solving the nonlinear equation is difficult, the linear equation is solvable:
$$\text{Instead of solving} \quad f(x) = 0 \quad \text{we solve} \quad p_1(x) = 0 \quad \text{at good guess } \bar{x}.$$
5. The produced output (if there were no problems) is then
$$\bar{x} - \big(f'(\bar{x})\big)^{-1} f(\bar{x})$$
6. This may or may not be a satisfactory output.

If we can positively answer to the following two questions:

1. Is the output a still good guess, so that we might try the procedure over again ?
2. If we repeat procedure, producing a sequence of good guesses
$$x_1 = \bar{x}, \quad x_{n+1} = \eta(x_n) = x_n - \big(f'(x_n)\big)^{-1} f(x_n),$$
is this sequence convergent ?

We will justify this convergence in theorem-proof style.

Convergence of Newton's method

The necessary conditions for the convergence of Newton's method are listed:

1. There exists a zero x_* of f with multiplicity $p \geq 1$, defined below.
2. f is C^1 in a neighborhood of x_*.

Multiplicity of zero

Definition 1. *We say x_* is a simple zero of f if*

$f \in C^1\big(I(r)\big)$ *for some* $r > 0$, $I(r) = (x_* - r, x_* + r)$ *and*
for $x \in I(r)$, *we can write* $\quad f(x) = c_1(x - x_*) + R(x) \quad$ *for some* $c_1 \neq 0$ *and*

$$\lim_{x \to x_*} \frac{|R(x)|}{|x - x_*|} = 0, \quad \lim_{x \to x_*} |R'(x)| = 0, \quad \textit{where the limit is taken in } I(r) \setminus \{x_*\}.$$

Definition 2. *We say x_* is a multiplicity p zero of f for $p > 1$ from the right if*

$f \in C^1\big(I(r)\big)$ *for some* $r > 0$, *where we denote* $I(r) = [x_*, x_* + r)$ *and*
for $x \in I(r)$, *we can write* $\quad f(x) = c_p(x - x_*)^p + R(x) \quad$ *for some* $c_p \neq 0$ *and*

$$\lim_{x \to x_*} \frac{|R(x)|}{|x - x_*|^p} = 0, \quad \lim_{x \to x_*} \frac{|R'(x)|}{|x - x_*|^{p-1}} = 0, \quad \textit{where the limit is taken in } I(r) \setminus \{x_*\}.$$

Remark 8.1. Multiplicify $p > 1$ zero from the left also can be similarly defined. Without loss of generallity we will only treat the case from the right.

Theorem 3. *Suppose x_* is either a simple zero of f or a multiplicity p zero from the right. Then, denoting the interval by the same symbole $I(r)$, there exists $0 < r' \leq r$ so that followings are true.*

1. *If $x \in I(r') \setminus x_*$ then $f'(x) \neq 0$.*
2. *If $x \in I(r') \setminus x_*$, define $\eta(x) = x - \frac{f(x)}{f'(x)}$. Then,*

$$\begin{aligned} &\textit{for some } 0 < \lambda < 1, \quad |\eta(x) - x_*| < \lambda |x - x_*|, \\ &\textit{in particular if } p > 1, \quad 0 < \eta(x) - x_* < \lambda(x - x_*). \end{aligned}$$

Remark 8.2. Because of the item 2, $\eta(x)$ is a member of $I(r')$. Hence, either $\eta(x)$ happen to be the solution x_* or the interation can be continued: If the initial good guess $x_0 \in I(r')$, the newton iteration either finds solution in finite steps or can be continued, actually to converge geometrically.

Proof. Let $p \geq 1$. We have that for $x \in I(r) \setminus \{x_*\}$

$$\begin{aligned}|f'(x)| = \left|pc_p(x-x_*)^{p-1} + R'(x)\right| &\geq |pc_p||x-x_*|^{p-1} - |R'(x)| \\ &= |x-x_*|^{p-1}\left(|pc_p| - \frac{|R'(x)|}{|x-x_*|^{p-1}}\right).\end{aligned}$$

By assumption, $\frac{|R'(x)|}{|x-x_*|^{p-1}} \to 0$ in $I(r) \setminus \{x_*\}$ as $x \to x_*$. Therefore, for $\epsilon = \frac{|pc_p|}{2} > 0$, there exists $0 < r'' \leq r$ such that

$$x \in I(r'') \setminus \{x_*\} \quad \Longrightarrow \quad \frac{|R'(x)|}{|x-x_*|^{p-1}} < \epsilon.$$

This implies that if $x \in I(r'') \setminus \{x_*\}$ then $f'(x) \neq 0$.

Now, assume $x \in I(r'') \setminus \{x_*\}$ and define $\eta(x) = x - \frac{f(x)}{f'(x)}$. Then,

$$\begin{aligned}\eta(x) - x_* &= x - x_* - \frac{f(x)}{f'(x)} \\ &= (x-x_*)\left(1 - \frac{f(x)}{f'(x)(x-x_*)}\right) = (x-x_*)\lambda(x).\end{aligned}$$

Now,

$$\begin{aligned}\lambda(x) &= 1 - \frac{c_p(x-x_*)^p + R(x)}{pc_p(x-x_*)^p + R'(x)(x-x_*)} \\ &= 1 - \frac{c_p + \frac{R(x)}{(x-x_*)^p}}{pc_p + \frac{R'(x)}{(x-x_*)^{p-1}}}.\end{aligned}$$

By assumption, $\lambda(x) \to 1 - \frac{1}{p}$ as $x \to x_*$ in $I(r'') \setminus \{x_*\}$. Therefore, for any $\epsilon > 0$, there exists $0 < r' \leq r''$ such that

$$x \in I(r') \setminus \{x_*\} \quad \Longrightarrow \quad \left|\lambda(x) - (1 - \frac{1}{p})\right| < \epsilon.$$

1. In case $p = 1$, the limit value $1 - \frac{1}{p} = 0$ and we choose $\epsilon = \frac{1}{2}$. Then we conclude

$$x \in I(r') \setminus \{x_*\} \quad \Longrightarrow \quad |\eta(x) - x_*| \leq |x - x_*|\ |\lambda(x)| < \frac{1}{2}|x - x_*|.$$

2. In case $p > 1$, the limit value $0 < 1 - \frac{1}{p} < 1$. Thus, there exists $t > 0$ such that

$$0 < 1 - \frac{1}{p} - t \quad \text{and} \quad 1 - \frac{1}{p} + t < 1.$$

   We choose $\epsilon = t$, then we conclude

$$x \in I(r') \setminus \{x_*\} \quad \Longrightarrow \quad 0 < \lambda(x)(x-x_*) = \eta(x) - x_* < \left(1 - \frac{1}{p} + t\right)(x - x_*).$$

□

1. We have proven that,
   (a) provided the initial guess is sufficiently close to the zero (neighborhood $I(r')$)
   (b) Newton's method generates a sequence that geometrically converges to the zero.
2. In reality, it is difficult to know in advance what value of r' would work, and what x_1 could be a good guess, since we do not know the solution x_*. So one would have to try many runs to achieve one's objectives.
3. For the method to converge, note that $f \in C^2$ is not necessary.

   Examples:

   (a) $f(x) = |x|^{1.5} + |x|^{1.8}$ has 0 as multiplicity 1.5 zero from right. Note that $f \notin C^2$ is still okay.
   (b) $f(x) = x + |x|^{1.8}$ has 0 as a simple zero, again $f \notin C^2$ is still okay.

   Counter Example: Newton's method may not work.

   (a) $f(x) = |x|^{1/3}$ does not satisfy the continuously differentiable assumption near 0.

Hypergeometric convergence

1. Newton's method, for $f \in C^2$ and for a simple zero, is known to converge much faster than geometric convergence, one can give a similar proof that the sequence converges hypergeomtrically,

$$\exists C > 0 \quad |\eta(x) - x_*| \leq C|x - x_*|^2.$$

   but we will not discuss further.

Solution error $|x - x_*|$ of muliplicity p zero

Testing error vs. Solution error

One should understand that we will have an implementation of this iteration

1. whose termination condition cannot be given by the solution error tolerance
$$|x_n - x_*| < \epsilon$$
simply because we do not know what x_* is.
2. Termination condition can be instead given by the testing error tolerance
$$|f(x_n)| < \eta.$$

Fractional order modulus of continuity

1. If we solve $f(x) = (x - x_*)^4$, knowing that
$$|x - x_*|^4 < 10^{-16} \quad \Longrightarrow \quad |x - x_*| < 10^{-4}.$$
2. Another way of putting this is that the local inverse valued in $x \geq x_*$
$$R : y \mapsto x_* + |y|^{\frac{1}{4}}$$
is continuous but is not as good as we may want: its modulus of continuity is of fractional order (bad).

The testing error tolerance cannot be arbitrarily small.

1. Of course, we are limited to having testing error tolerance, that of the machine,
$$\eta_{machine} \sim 10^{-16}$$
2. Deeper thinking: it is fair we also consider the confidence level on the function f, say if it has some coefficients or parameters with the certain level of confidence.

   Extremely speaking, if f evaluation involves numbers, say 4 significant digits with number scale 10^5, testing error tolerance less than 1 will make no sense.

Conclusion of fractional order modulus of continuity

1. Newton's method, under C^1 assumption and the presence of a good guess, is known to converge to the zero, regardlessly of multiplicity, quite fast: geometrically or hypergeometrically.
2. Since there is a limit for the testing error tolerance, due to the machine limit or the confidence level of f evaluation itself, the geometric or hypergeometric convergence of Newton's method only has a meaning up to the point it reaches the testing error tolerance.
3. If multiplicity p is high, then the solution error may not be as small as we want, while the testing error is as small as tolerance allows.

Chapter 9

Consistency, Convergence, and Errors

Many problems can be formulated as equation solving problems, and equation solving problem can be analyzed, by the least, in a certain systematic way.

To take an example, we consider the problem of orthogonalization of an $n \times n$ square matrix:

For given $A \in R^{n\times n}$, find the orthogonal matrix $O \in \mathbb{R}^{n\times n}$, and the upper triangular matrix $R \in \mathbb{R}^{n\times n}$ such that $A = OR$.

1. We let $X = X_1 \times X_2$, X_1 the set of all $n \times n$ orthogonal matrices, X_2 the set of all $n \times n$ upper triangular matrices.
2. We let Y be the set of all $n \times n$ matrices.
3. We let $F(O, R) = OR$.

and we solve $F(O, R) = A$ for given A.

X, Y, and $F : X \to Y$

- X and Y are metric spaces, and F is assumed to be continuous and surjective.
- The testing whether $F(x)$ equals to y has to be checkable accurate enough.
- In considering the problem to solve $F(x) = y$, the existence theory is from upstream, i.e., somebody else other than numerical analyst has done it.

Below, we use notation $|x - x'|$ for the distance, adapted to the normed space, but you can adjust this to the distance function in metric space.

(Parametrized) Modified problem $\bar{F}_j$

- The objective is to solve the equation $F(x) = y$, by taking the limit of a parametrized solvable problems $\bar{F}_j$.

  In general,

  1. Let X_j be a subset of X,
  2. Let Y_j be a subset of Y,
  3. The modified problem we consider is between X_j and Y_j:

$$\bar{F}_j : X_j \to Y_j.$$

(Parametrized) Approximate Solver $\bar{R}_j$

- The solver of the solvable problem $\bar{F}_j(x_j) = y_j$ is denoted by $\bar{R}_j : Y_j \to X_j$,

$$\bar{F}_j\big(\bar{R}_j(y_j)\big) = y_j.$$

- For input $y \in Y_j$, $\bar{R}_j$ produces one output.

  This means, if there are many solutions, the step of deciding $\bar{F}_j$ and $\bar{R}_j$ includes making the choice among solutions. This may or may not be explicit in the approximate algorighm $\bar{R}_j$.

  From now on, we assume $\bar{R}_j$ is bijective between Y_j and its range.

The consistency is the notion saying we are not doing irrelevant to the original problem. There are two kinds of consistency.

Consistency of $\bar{F}_j$

Definition 1. *We say $\bar{F}_j$ is η_j-consistent at $x \in X$ if*

$$|F(x) - \bar{F}_j(x)|_Y < \eta_j.$$

We call $|F(x) - \bar{F}_j(x)|_Y$ the testing error at $x \in X$.

Consistency of $\bar{R}_j$

Definition 2. *We say $\bar{R}_j$ is η_j-consistent at $y \in Y$ if*

1. *$\bar{R}_j(y) \in X$, and*
2. *$\left|F\big(\bar{R}_j(y)\big) - y\right|_Y < \eta_j$.*

We call $\left|F\big(\bar{R}_j(y)\big) - y\right|_Y$ the testing error at $y \in Y$.

1. Consistency is the least requirement, if a method is not consistent in any of two kinds, the method must be abandoned. If one is a developer for a certain new problem, one should try to build a consistent method.
2. Problemwisely, and purposewisely, we may or may not want to go further than having a consistent algorithm.

Newton's method iteration from a good guess for C^1 function, is a consistent algorithm.

Orthogonalization algorithms are good examples:

- Gram-Schmidt algorithm $\tilde{R}_{GS}$ to get (O, R) out of A is inconsistent if A is rank deficient.

  The algorithm gives me O and R such that $OR \simeq A$ but O is far from being orthogonal matrix.

- Householder algorithm $\tilde{R}_H$ to get (O, R) out of A is consistent. However, it lacks the solution error control. If A' is say small perturbation of A, then the algorithm gives me orthogonal O' and upper triangular R' and $O'R' \simeq A'$ but O and O' is not at all close to each other.

However, the Householder algorithm is used everywhere for the right purposes.

Solution Error

Accuracy of $\bar{R}_j$

Definition 3. *We say $\bar{R}_j$ is ϵ_j-accurate if*

$$\exists x_* \in X \quad \textit{such that} \quad F(x_*) = y_*, \quad |x_* - \bar{R}_j(y_*)|_X < \epsilon_j.$$

Definition 4. *We say $(\bar{R}_j)$ is a convergent algorithm if*

$$\exists x_* \in X \quad \textit{such that} \quad F(x_*) = y_*, \quad |x_* - \bar{R}_j(y_*)|_X < \epsilon_j, \quad \epsilon_j \to 0 \quad \textit{as } j \to \infty.$$

A Consistent algorithm is upgraded to an Accurate algorithm, in case true solver or approximate solver satisfy a certain continuity.

1. The first leads to the a posteriari solution error analysis,
2. The latter leads to the a priori solution error analysis.

1. a posteriari solution error analysis

Continuity of true local inverse of F

In the example $F(x) = (x - x_*)^4$, one can take

$$R(y) = x_* + y^{1/4} \quad y \geq 0$$

taking values in $x \geq x_*$. Then R is continuous but its modulus continuity is of fractional order.

Let $\bar{y} \in Y$ be fixed, and suppose $F(\bar{x}) = \bar{y}$. We consider the following statement for F at some $\bar{y}$:

Statement: for every $\epsilon > 0$, there exists $\eta > 0$ such that

$$|y - \bar{y}|_Y < \eta \text{ implies there exists } x \in X \text{ with } F(x) = y \text{ and } |x - \bar{x}| < \epsilon.$$

We denote the map picking up one such existing element by a local true inverse R defined near $\bar{y}$. We assume $\eta(\epsilon, \bar{y})$ is monotone in ϵ and known to us. Then we let its inverse $\epsilon(\eta, \bar{y})$ be the modulus of continuity of local true inverse R near $\bar{y}$.

Assume consistency of $\bar{R}_j$ at y_* + continuity statement at $F\big(\bar{R}_j(y_*)\big)$:

1. Suppose $(\bar{R}_j)$ is consistent: $(\bar{R}_j)$ is η_j-consistent at y_* and $\eta_j \to 0$ as $j \to \infty$.
2. Suppose above statement is true at $\bar{y} = F\big(\bar{R}_j(y_*)\big)$ and the modulus of continuity

$$\epsilon(\eta, \bar{y})$$

   is known at $\bar{y}$.

Conclusion:

Let y_* be given and fix some j_0. We run the algorithm $\bar{R}_{j_0}$ and compute the following.

1. We compute $\bar{y} = F\big(R_{j_0}(y_*)\big)$. By consistency, we know that

$$|\bar{y} - y_*|_Y < \eta_{j_0}(y_*).$$

2. From the statement and modulus of continuity at $\bar{y}$

$$\text{There exists } x_* \in X \text{ with } F(x_*) = y_* \text{ and } \big|x_* - R_j(y_*)\big|_X < \epsilon(\eta_{j_0}(y_*)).$$

Since the testing error $|\bar{y} - y_*|_Y$ and its corresponding solution error are quantities available after running $\bar{R}_{j_0}$, this is known as a posteriari error analysis.

2. a priori solution error analysis

Continuity of whole approximate inverses (R_j)

Instead of R, we work with continuity of approximate inverses (R_j) but as a whole.

Let $\bar{y} \in Y$ be fixed, and suppose $F(\bar{x}) = \bar{y}$. We consider the following statement for (R_j) at some $\bar{y}$:

Statement: for every $\epsilon > 0$, there exists $\eta > 0$ such that

$$|y - \bar{y}|_Y < \eta \text{ implies that for every } j,\ |\bar{R}_j(y) - \bar{R}_j(\bar{y})| < \epsilon.$$

(This is known as the notion of equi-continuity at $\bar{y}$ for a sequence of functions.)

We assume $\eta(\epsilon, \bar{y})$ is monotone in ϵ and known to us. Then we let its inverse $\epsilon(\eta, \bar{y})$ be the modulus of continuity of the $(\bar{R}_j)$ near $\bar{y}$.

Assume consistency of $\bar{F}_j$ at x_* + continuity statement at y_*.

1. Suppose $(\bar{F}_j)$ is consistent: (F_j) is η_j-consistent at x_* and $\eta_j \to 0$ as $j \to \infty$.
2. Suppose above statement is true at y_* and the modulus of continuity

   $$\epsilon(\eta, y_*)$$

   is known at y_*.

Conclusion:

1. We let $y_j = F_j(x_*)$. By consistency, we *a priori* know that at unknown x_*

   $$|\bar{F}_j(x_*) - F(x_*)|_Y = |y_j - y_*|_Y < \eta_j(x_*).$$

2. Let $\bar{R}_j(y_*) = x_j$. From the statement and modulus of continuity at y_*, we *a priori* know that

   $$\left|\bar{R}_j(y_j) - \bar{R}_j(y_*)\right|_X < \epsilon(\eta_j(x_*))$$

   i.e.,

   $$|x_* - x_j|_X < \epsilon(\eta_j(x_*)).$$

1. In a priori error analysis,
   (a) We do not know what x_* is. Therefore, the consistency level $\eta(x_*)$, which depends on x, assumes the worst case scenario in X.
   (b) It is us to implement the algorithm $\bar{R}_j$, the modulus of continuity $\epsilon(\eta, y_*)$ is in principle given to us.
   (c) The solution error level $\epsilon(\eta, y_*)$ is the error that is given in advance we run the algorithm.
2. In a posteriari error analysis,
   (a) We do know what y_* is, and after we run the algorithm we know what $\bar{y} = F\big(\bar{R}_j(y_*)\big)$ is. Therefore, the consistency level $\eta(\bar{y})$ assumes the adapted scenario that suffices to work only in the neighborhood of $\bar{y}$.
   (b) It is the problem that decides the modulus of continuity $\epsilon(\eta, \bar{y})$ for true R, and has to be known to us by the upstream theoretical analysis.
   (c) The solution error level $\epsilon(\eta, \tilde{y})$ is the error that is given after we run the algorithm.

Chapter 10

Vector space of pp functions

1. In 1d, we recall that the set of all polynomials of order at most n on $[a, b]$ is the vector space $\mathcal{P}^{<n+1}([a, b])$ of dimensions $n + 1$.
2. In the similar way, we consider the set of piecewise polynomials on $[a, b]$ in $1d$.

A few notions:

1. We let $[a, b] = I$ be the domain of pp function. We call I the global domain.
2. We partition the global domain into L local intervals.
3. We consider its partition by *break points*
$$a = t_1 < t_2 < t_3 < \cdots < t_{L+1} = b.$$
4. This partition induces local intervals
$$I_\ell = [t_\ell, t_{\ell+1}], \quad \ell = 1, 2, \cdots, L.$$
5. *Interfaces* are intersections of such local intervals. Here interfaces are simply
$$(t_i)_{i=2}^{L}.$$

Assembling a pp function φ by polynomials on I_ℓ.

The first example

1. Suppose for each local interval I_ℓ, the maximum order of polynomial piece of $\varphi\big|_{I_\ell}$ on I_ℓ we want is n_ℓ.
2. Choose $p_\ell \in \mathcal{P}^{<n_\ell+1}(I_\ell)$, for each $\ell = 1, 2, \cdots, L$. This defines a pp function φ:
$$\varphi(x) = p_\ell(x) \quad \text{if } x \in \operatorname{int}(I_\ell).$$
3. If $\varphi(x)$ and $\psi(x)$ are two such functions, then so is $\alpha\varphi(x) + \beta\psi(x)$ for $\alpha, \beta \in \mathbb{R}$. We collect all such pp functions to form a vector space denoted by
$$\Pi\Big([a, b]\ ;\ (t_i)\ ;\ (n_\ell)\Big), \quad \text{where} \quad (t_i) = (t_i)_{i=1}^{L+1}, \quad (n_\ell) = (n_\ell)_{\ell=1}^{L}.$$
If there is no confusion, we will also simply use $\Pi([a, b])$.
4. Total degrees of freedom N, or the dimensions of $\Pi([a, b])$ is
$$N = n_1 + 1 + n_2 + 1 + \cdots + n_L + 1.$$

Remark 10.1. We are careless if the local domain is $(t_{\ell-1}, t_\ell)$ or $[t_{\ell-1}, t_\ell)$ or $[t_{\ell-1}, t_\ell]$ because in I_ℓ, $p_\ell(x)$ gives rise to the sided limit from inside, i.e., p_ℓ is uniformly continuous on I_ℓ.

pp functions with matching conditions at interfaces

1. On top of previous considerations, suppose we want to impose matching conditions on each interface t_i $k = 2, 3, \cdots, L$.
2. Let us restrict ourselves to the following types of matching conditions.
   (a) At each t_i, $i = 2, 3, \cdots, L$ we decide how many matching conditions we want to impose, that are nonnegative integers
   $$m_i, \quad i = 2, 3, \cdots, L.$$
   (b) If $m_i = 0$, then no condition is imposed at t_i. If $m_i \geq 1$, it means that
   $$\begin{aligned} \varphi(t_i-) &= \varphi(t_i+) \\ \varphi'(t_i-) &= \varphi'(t_i+) \\ &\vdots \\ \frac{d^{m-1}\varphi}{dx^{m-1}}(t_i-) &= \frac{d^{m-1}\varphi}{dx^{m-1}}(t_i+) \end{aligned}$$
   up to derivatives of order $m_i - 1$ for each interface t_i.
   (c) The number of matching conditions m_i is assumed to not exceed the polynomial orders of interface's left and right intervals.

1. If $\varphi(x)$ and $\psi(x)$ are two such functions, then so is $\alpha\varphi(x) + \beta\psi(x)$ for $\alpha, \beta \in \mathbb{R}$. We collect all such pp functions to form a vector space denoted by
   $$\Pi\Big([a, b]\,;\,(t_i)\,;\,(n_\ell)\,;\,(m_i)\Big), \quad \text{where} \quad (t_i) = (t_i)_{i=1}^{L+1}, \quad (n_\ell) = (n_\ell)_{\ell=1}^{L}, \quad (m_i) = (m_i)_{i=2}^{L}.$$
   If there is no confusion, we will also simply use $\Pi([a, b])$.
2. If there are no imposed matching conditions, the total degrees of freedom N was
   $$N = n_1 + 1 + n_2 + 1 + \cdots + n_L + 1.$$
3. Since we imposed constraints in total
   $$\sum_{i=2}^{L} m_i = M,$$
   degrees of freedom are reduced by M, and thus the dimensions of $\Pi([a, b])$ is
   $$N - M.$$

Piecewise linear and continuous function

Splines of order r

1. Particularly handy choice of $\Pi[a,b]$ is the case where for $m \in \{0\} \cup \mathbb{N}$

$$n_1 = n_2 = \cdots = n_L = m, \quad m_2 = m_3 = \cdots = m_L = m, \quad .$$

   (a) $m = 0$: piecewise constant function without matching conditions.
   (b) $m = 1$: piecewise linear function that is continuous,
   (c) $m = 2$: piecewise quadratic that is C^1 function,
   (d) $m = 3$: piecewise cubic that is C^2 function,

$$\vdots$$

2. We denote such spaces for each m by $\mathsf{S}^{<r}([a,b])$ with $r = m+1$. A member of $\mathsf{S}^{<r}([a,b])$ is called a spline function of (spline) order r.

3. The dimensions of $\mathsf{S}^{<r}([a,b])$ is computed by

$$N-M = \Big(\sum_{\ell=1}^{L} n_\ell+1\Big)-\Big(\sum_{i=2}^{L} m_i\Big) = \Big(\sum_{\ell=1}^{L} r\Big)-\Big(\sum_{i=2}^{L} r-1\Big) = rL-(r-1)(L-1) = L+r-1.$$

Remark 10.2. The vector spaces $\mathsf{S}^{<r}([a,b])$ for $r = 1, 2, \cdots$ has the nice property:

Let $a = t_1 < t_2 < \cdots < t_{L+1} = b$ be the break points, which are shared. Then,

$$\mathsf{S}^{<1}([a,b]),\ \mathsf{S}^{<2}([a,b]),\ \mathsf{S}^{<3}([a,b]), \cdots$$

are respectively subsets of

$$B([a,b]),\ C^0([a,b]),\ C^1([a,b]), \cdots$$

and one can check that the derivative of r spline becomes $r-1$ spline for $r \geq 3$.

Splines of order r on $\mathbb{R}$

1. For later purposes, in addition to the space $\mathsf{S}^{<r}([a,b])$ of r-splines on $[a,b]$, we introduce the space
$$\mathsf{S}_c^{<r}(\mathbb{R})$$
defined on the whole domain $\mathbb{R}$ with compact support.

2. Suppose
$$-\infty < t_1 < t_2 < \cdots < t_k < \infty$$
for some $k > r$ induces partition of $\mathbb{R}$:
$$(-\infty, t_1],\ [t_1, t_2],\ [t_2, t_3],\ \cdots,\ [t_{k-1}, t_k],\ [t_k, \infty)$$

3. We agree on that the polynomial for the far field
$$(-\infty, t_1] \quad \text{and} \quad [t_k, \infty)$$
is the zero function.

4. Joined from the zero function on the far field, we consider a pp function φ whose polynomial order and number of matching conditions are
$$m = r - 1.$$

5. We denote the space by $\mathsf{S}_c^{<r}(\mathbb{R})$ of r-splines on $\mathbb{R}$ with compact support.

6. The dimensions of $\mathsf{S}_c^{<r}(\mathbb{R})$ is computed by
$$N - M = r(k-1) - (r-1)k = k - r.$$

This is an additional material on placing break points on $\mathbb{R}$.

Placing break points

We make rules for break points to define a r-spline on the whole $\mathbb{R}$.

1. Initially we begin with no break point. Then the whole $\mathbb{R}$ is taken by one polynomial of order at most $r-1$.

   In other words,

   - at any x_0 that is not a break point, the point assumes r constraints,

   $$\begin{aligned} \varphi(x_0-) &= \varphi(x_0+) \\ \varphi'(x_0-) &= \varphi'(x_0+) \\ &\vdots \\ \frac{d^m\varphi}{dx^m}(x_0-) &= \frac{d^m\varphi}{dx^m}(x_0+) \end{aligned}$$

   where $m = r-1$.
   - Polynomials p and q of order at most $r-1$ with r matching conditions must coincide.
   - Hence, crossing x_0, polynomial keeps remain same.

2. If a break point is placed at t, then it breaks one matching condition there, the last one:

   $$\frac{d^m\varphi}{dx^m}(t-) \neq \frac{d^m\varphi}{dx^m}(t+)$$

   - Let $(-\infty, t)$ be taken by a polynomial p of order at most $r-1$.
   - Crossing t, one degree of freedom is released from p, in choosing a polynomial q on (t, ∞).

3. As we introduce a few break points t_1, t_2, $\cdots$, t_k, it breaks precisely one matching condition (the last one) at each of the points.

4. This is going to be quite general, in the sense that we can place another break point over the pre-existing break point.

   - If break points are placed twice at t, then it breaks the last two matching conditions.
   - If break points are placed r times at t, then it breaks all r matching conditions, so no condition is imposed at t, and jump discontinuity is allowed crossing t.

As one introduce finitely many break points

$$t_1 \leq t_2 \leq t_3 \leq \cdots \leq t_k$$

this can be arranged in the way

$$\tau_1 < \tau_2 < \cdots < \tau_K$$

and each τ_i is attached the multiplicity of break points and $\mathbb{R}$ is partitioned into

$$(-\infty, \tau_1],\ [\tau_1, \tau_2],\ [\tau_2, \tau_3],\ \cdots,\ [\tau_{K-1}, \tau_K],\ [\tau_K, \infty)$$

Exercise

Keep the same number of break points k but let a few break points coalesce. Show that the dimensions $k - r$ remains same.

Chapter 11

Bump-shaped splines as Basis elements

1. We are going to assemble bump-shaped r-splines, which are greatly useful. In particular, these will be used as basis elements for $\mathsf{S}^{<r}([a,b])$.
2. Those bump-shaped r-splines, firstly are introduced as basis elements of

$$\mathsf{S}_c^{<r}(\mathbb{R})$$

defined on the whole domain $\mathbb{R}$, which comes more natural.

The followings are the kinds of r-bumps we consider:

Example: $r = 1$

Example: $r = 2$

Example: $r = 3$

The function $x \mapsto B(x; i; r)$

Suppose

$$-\infty < t_1 < t_2 < \cdots < t_k < \infty$$

for some $k > r$ induces partition of $\mathbb{R}$.

We define $k - r$ distinct r-bumps $B(x; i; r) \in \mathsf{S}_c^{<r}(\mathbb{R})$ for $i = 1, 2, \cdots, k - r$,

a bump starting at t_i and ending at t_{i+r}.

Some remarks

1. As same as other functions who have names such as

   $$\sin(x), \quad \exp(x), \quad Relu(x), \quad \cdots$$

   $B(x; i; r)$ functions also should be regarded as a pre-existing function with name and used as so.

2. $B(x; i; r)$ functions are called b-splines, where b stands for the basis from original textbook.

3. There are many different ways to define b-spline functions and b-splines can be interpreted from many different aspects. They enjoy rich and interesting properties.

4. We will follow one specific way of definition, and one can see other in the textbook

   Carl de Boor, A practical guide to splines

   where break points coalesce is allowed. In spline context, break points are also called *knots*.

We give definitions by induction.

- $r = 1$. We define $k-1$ b-splines:

  For each $i = 1, 2, \cdots, k-1$, define

$$B(x; i; 1) = \chi_{(t_i, t_{i+1})}(x) = \begin{cases} 1 & \text{if } x \in (t_i, t_{i+1}) \\ 0 & \text{otherwise} \end{cases}$$

- $r = 2$. We define $k-2$ b-splines:

  For each $i = 1, 2, \cdots, k-2$, we do the following:

  1. We first define the normalization of 1-bumps,

$$C(x; i; 1) = \frac{B(x; i; 1)}{\int_{\mathbb{R}} B(i; x'; 1)\, dx'} \left(= \frac{B(x; i; 1)}{\int_{t_i}^{t_{i+1}} B(i; x'; 1)\, dx'} = \frac{B(i; x; 1)}{t_{i+1} - t_i} \right).$$

  2. Then, $C(x; i; 1) - C(x; i+1; 1)$ is a 1-spline with zero integral supported in

$$[t_i, t_{i+2}].$$

  3. We define

$$B(x; i; 2) = \int_{-\infty}^{x} C(x'; i; 1) - C(x'; i+1; 1)\, dx'.$$

- If b-splines of order $1, 2, \cdots, r-1$ are defined we define $k-r$ b-splines:

  For each $i = 1, 2, \cdots, k-r$, we do the following:

  1. We define the normalization of $(r-1)$-bumps,

$$C(x; i; r-1) = \frac{B(x; i; r-1)}{\int_{\mathbb{R}} B(x'; i; r-1)\, dx'} = \frac{B(x; i; r-1)}{\int_{t_i}^{t_{i+r-1}} B(x'; i; r-1)\, dx'}.$$

  2. Then, $C(x; i; r-1) - C(x; i+1; r-1)$ is a $(r-1)$-spline with zero integral supported in

$$[t_i, t_{i+r}].$$

  3. We define

$$B(x; i; r) = \int_{-\infty}^{x} C(x'; i; r-1) - C(x'; i+1; r-1)\, dx'.$$

Example of direct calculations

$B(x;i;r)$ is r-bumps

- $B(x;i;r) \in \mathsf{S}_c^{<r}(\mathbb{R})$

  Indeed,

  1. Obviously $B(x;i;1) \in \mathsf{S}_c^{<1}(\mathbb{R})$.
  2. Since $C(x;i;1) - C(x;i+1;1)$ is a bounded piecewise constant function, by FTC
  $$B(x;i;2) = \int_{-\infty}^{x} C(x';i;1) - C(x';i+1;1)\, dx'$$
  is continuous function.
  3. At every interval $[t_j, t_{j+1}]$, and for $x \in [t_j, t_{j+1}]$
  $$B(x;i;2)\Big|_{[t_j,t_{j+1}]} = \int_{-\infty}^{t_j} C(x';i;1) - C(x';i+1;1)dx' + \int_{t_j}^{x} C(x';i;1) - C(x';i+1;1)dx'$$
  is a linear function on the interval. Thus, $B(x;i;2) \in \mathsf{S}_c^{<2}(\mathbb{R})$.
  4. Similarly, it is straightforward to check $B(x;i;3)$ is a C^1 function and piecewise quadratic.
  5. Inductively, we observe $B(x;i;r) \in \mathsf{S}_c^{<r}(\mathbb{R})$.

- $B(x;i;r)$ is nonnegative and compactly supported in
$$[t_i, t_{i+r}].$$

$B(x;i;r)$ as basis elements

- $\mathsf{span}\left\langle B(x;1;r), B(x;2;r), \cdots, B(x;k-r;r)\right\rangle = \mathsf{S}_c^{<r}(\mathbb{R})$

  That $B(x;1;r), B(x;2;r), \cdots, B(x;k-r;r)$ are $k-r$ linearly independent elements can be checked from the fact that the supports of b-splines
  $$[t_i, t_{i+r}]$$
  are not mutually inclusive.

- $\mathsf{span}\left\langle B(x;1;r)\big|_{[a,b]},\ B(x;2;r)\big|_{[a,b]},\ \cdots,\ B(x;L+r-1;r)\big|_{[a,b]}\right\rangle = \mathsf{S}^{<r}([a,b])$.

  1. Now, let us take $k = L+2r-1$ for some positive integer L. We are considering the numbering such that
  $$-\infty < t_1 < t_2 < \cdots < t_r < t_{r+1} < \cdots < t_{L+r} < \cdots < t_{L+2r-1} < \infty$$
  $$a = \tau_1 < \tau_2 < \cdots < \tau_{L+1} = b$$
  with $\tau_1 = t_r, \quad \tau_2 = t_{r+1}, \quad \cdots, \quad \tau_{L+1} = t_{L+r},$

  i.e., the finite sequence $(\tau_\ell)_{\ell=1}^{L+1}$ is embedded in the middle of sequence $(t_i)_{i=1}^{L+2r-1}$ so that $r-1$ break points are appended before τ_1 and after τ_{L+1} respectively.

2. Then,

$$\mathsf{span}\left\langle B(x;1;r)\big|_{[a,b]},\ B(x;2;r)\big|_{[a,b]},\ \cdots,\ B(x;L+r-1;r)\big|_{[a,b]}\right\rangle = \mathsf{S}^{<r}\Big([a,b]\ ;\ (\tau_\ell)\Big).$$

Partition of unity

One another property that are not readily seen is that

$$\text{for every } x \in [a,b], \quad B(x;1;r) + B(x;2;r) + \cdots + B(x;L+r-1;r) = 1.$$

This is because of the following.

For $x \in [a,b]$,

$$\begin{aligned}
&\sum_{i=1}^{L+r-1} B(x;i;r) \\
&= \int_{-\infty}^{x} \Big(C(x';1;r-1) - C(x';2;r-1)\Big) + \Big(C(x';2;r-1) - C(x';3;r-1)\Big) + \cdots \\
&\qquad + \Big(C(x';L+r-1;r-1) - C(x';L+r;r-1)\Big)\, dx' \\
&= \int_{-\infty}^{x} \Big(C(x';1;r-1) - C(x';L+r;r-1)\Big)\, dx'
\end{aligned}$$

(since $x \geq a = t_r$ and $x \leq b = t_{L+r}$)

$$= \int_{-\infty}^{t_r} C(x';1;r-1)(x')\, dx' = 1.$$

Remark 11.1. It is true that the choice of $t_1, t_2, \cdots, t_{r-1}$ and $t_{L+r+1}, t_{L+r+2}, \cdots, t_{L+2r-1}$, or the choice of appended $2(r-1)$ break points affect the definition of some b-splines. One can even choose $t_1 = t_2 = \cdots = t_{r-1} = t_r$ and $t_{L+r} = t_{L+r+1} = t_{L+r+2} = \cdots = t_{L+2r-1}$ for the generalized definition. Those who are interested in are referred to the textbook of Carl de Boor.

Chapter 12

Interpolation by $\mathsf{S}^{<r}([a,b])$

1. For a given sequence $(t_i)_{i=1}^{L+1}$ of break points, the vector space $\mathsf{S}^{<r}([a,b])$ of dimensions $L+r-1$ has been defined.

2. Suppose also that we are given sampling data

$$\big(x_1, f(x_1)\big),\ ,\big(x_2, f(x_2)\big),\ \cdots,\ \big(x_N, f(x_N)\big)$$

   for some N. If the number of sampling data and the dimensions of the vector space are same, then one pp function can be picked up as an interpolation of the sampling data.

Let us examine the piecewise constant interpolation, piecewise linear interpolation, and piecewise cubic interpolation

In this chapter, we assume that the sampling points are all distinct:

We let $x_1, x_2, \cdots, x_{n+1} \in [a,b]$ and also

$$a \le x_1 < x_2 < x_3 < \cdots < x_N \le b.$$

and the sampling data is from a smooth function f.

piecewise constant interpolation ($r = 1$)

1. We choose break points

$$a = t_1 < t_2 < \cdots < t_N < t_{N+1} = b$$

   in such a way that every interval $[t_\ell, t_{\ell+1}]$ contains precisely one sampling point x_ℓ.

2. This corresponds the case with $L + 1 = N + 1$, and

$$\dim \mathsf{S}^{<1}([\mathsf{a},\mathsf{b}]) = L + r - 1 = N + 1 - 1 = N.$$

piecewise linear interpolation ($r = 2$)

1. We can choose break points = sampling points

$$t_i = x_i, \quad i = 1, 2, \cdots, N.$$

2. This corresponds the case with $L + 1 = N$, and

$$\dim \mathsf{S}^{<2}([\mathsf{a},\mathsf{b}]) = L + r - 1 = N - 1 + 2 - 1 = N.$$

piecewise cubic interpolation ($r = 4$)

Now, we discuss the cubic spline interpolation of the case $r = 4$.

There are three different well-known cubic spline interpolations.

1. 'Not a knot' cubic spline

1. For the 'Not a knot' cubic spline, we choose break points

$$\hat{t}_1 = x_1, \quad \hat{t}_2 = x_3, \quad \hat{t}_3 = x_4, \quad \cdots \quad \hat{t}_{N-3} = x_{N-2}, \quad \hat{t}_{N-2} = x_N,$$

   i.e., the second point x_2 and the second last poinst x_{N-1} are not *knots*.

2. This corresponds the case with $L + 1 = N - 2$, and the dimensions of the vector space $\mathsf{S}^{<4}([\mathsf{a}, \mathsf{b}])$

$$\dim \mathsf{S}^{<4}([\mathsf{a}, \mathsf{b}]) = L + r - 1 = N - 3 + 4 - 1 = N.$$

3. This choice is 'good' in the sense that the curve ends without the inflection point in the first and the last interval.

4. There is a classical way of computing piecewise polynomials, but here we simply use b-splines. Let us embed $(\hat{t}_i)_{i=1}^{N-2}$ into $(t_i)_{i=1}^{N+4}$

$$-\infty < t_1 < t_2 < t_3 < a \le t_4 < t_5 < \cdots < t_{N+1} \le b < t_{N+2} < t_{N+3} < t_{N+4} < \infty$$
$$t_4 = \hat{t}_1, \quad t_5 = \hat{t}_2, \quad \cdots, \quad t_{N+1} = \hat{t}_{N-2}$$

   so that $B(x; i; 4)\big|_{[a,b]}$ for $i = 1, 2, \cdots, N$ forms a basis. In particular, let us write

$$b_i(x) = B(x; i; 4) \quad i = 1, 2, \cdots, N$$

   for short.

5. We look for an element in the form

$$\varphi = \sum_{i=1}^{N} c_i b_i(x)$$

   satisfying

$$\varphi(x_j) = f(x_j), \quad j = 1, 2, \cdots, N$$

6. We solve the system

$$\begin{pmatrix} b_1(x_1) & b_2(x_1) & \cdots & b_N(x_1) \\ b_1(x_2) & b_2(x_2) & \cdots & b_N(x_2) \\ b_1(x_3) & b_2(x_3) & \cdots & b_N(x_3) \\ \vdots & \vdots & \vdots & \vdots \\ b_1(x_N) & b_2(x_N) & \cdots & b_N(x_N) \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ \vdots \\ c_N \end{pmatrix} = \begin{pmatrix} f(x_1) \\ f(x_2) \\ f(x_2) \\ \vdots \\ f(x_N) \end{pmatrix}.$$

7. Note that, since $B(x;i;4)$ is supported in $[t_i, t_{t+4}]$, other than
$$B(t_{i+1};i;4), \quad B(t_{i+2};i;4), \quad B(t_{i+3};i;4)$$
all other values at break points are zero. Also, $B(x;i;4)$ is bumb-shaped with the peak around t_{i+2}. Thus the matrix is banded and diagonal dominant.

2. Natural cubic spline

1. For the natrual cubic spline, we choose break points = sampling points
$$\hat{t}_i = x_i, \quad i = 1, 2, \cdots, N.$$
2. This corresponds the case with $L+1 = N$, and the dimensions of the vector space $\mathsf{S}^{<4}([\mathsf{a},\mathsf{b}])$
$$\dim \mathsf{S}^{<4}([\mathsf{a},\mathsf{b}]) = L + r - 1 = N - 1 + 4 - 1 = N + 2.$$
which is greater than N by two.
3. To fix an element in $\mathsf{S}^{<4}([\mathsf{a},\mathsf{b}])$, we force the pp function to satisfy at the first and the last point
$$\varphi''(x_1) = \varphi''(x_N) = 0.$$
4. This choice is 'good' in the sense that the curve ends without the inflection point in the first and the last interval.
5. There is a classical way of computing piecewise polynomials, but here we simply use b-splines. Let us embed $(\hat{t}_i)_{i=1}^N$ into $(t_i)_{i=1}^{N+6}$
$$-\infty < t_1 < t_2 < t_3 < a \le t_4 < t_5 < \cdots < t_{N+3} \le b < t_{N+4} < t_{N+5} < t_{N+6} < \infty$$
$$t_4 = \hat{t}_1, \quad t_5 = \hat{t}_2, \quad \cdots, \quad t_{N+3} = \hat{t}_N.$$
so that $B(x;i;4)\big|_{[a,b]}$ for $i = 1, 2, \cdots, N+2$ forms a basis. In particular, let us write
$$b_i(x) = B(x;i;4) \quad i = 1, 2, \cdots, N+2$$
for short.
6. We look for an element in the form
$$\varphi = \sum_{i=1}^{N+2} c_i b_i(x)$$
satisfying
$$\varphi(x_j) = f(x_j), \quad j = 1, 2, \cdots, N$$
$$\varphi''(x_1) = \varphi''(x_N) = 0.$$
7. We solve the system
$$\begin{pmatrix} b_1''(x_1) & b_2''(x_1) & \cdots & b_{N+2}''(x_1) \\ b_1(x_1) & b_2(x_1) & \cdots & b_{N+2}(x_1) \\ b_1(x_2) & b_2(x_2) & \cdots & b_{N+2}(x_2) \\ b_1(x_3) & b_2(x_3) & \cdots & b_{N+2}(x_3) \\ \vdots & \vdots & \vdots & \vdots \\ b_1(x_N) & b_2(x_N) & \cdots & b_{N+2}(x_N) \\ b_1''(x_N) & b_2''(x_N) & \cdots & b_{N+2}''(x_N) \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ \vdots \\ c_N \\ c_{N+1} \\ c_{N+2} \end{pmatrix} = \begin{pmatrix} 0 \\ f(x_1) \\ f(x_2) \\ f(x_2) \\ \vdots \\ f(x_N) \\ 0 \end{pmatrix}.$$

3. Clamped cubic spline

1. For the clamped cubic spline, we assume two more sampling data:

$$f''(x_1), f(x_1), f(x_2), f(x_3), \cdots, f(x_N), f'(x_N).$$

2. With the same choice of break points as in the natural cubic spline, we look for an element in the form

$$\varphi = \sum_{i=1}^{N+2} c_i b_i(x)$$

satisfying

$$\begin{aligned} \varphi'(x_1) &= f'(x_1) \\ \varphi(x_j) &= f(x_j), \quad j = 1, 2, \cdots, N \\ \varphi'(x_N) &= f'(x_N) \end{aligned}$$

3. We solve the system

$$\begin{pmatrix} b_1'(x_1) & b_2'(x_1) & \cdots & b_{N+2}'(x_1) \\ b_1(x_1) & b_2(x_1) & \cdots & b_{N+2}(x_1) \\ b_1(x_2) & b_2(x_2) & \cdots & b_{N+2}(x_2) \\ b_1(x_3) & b_2(x_3) & \cdots & b_{N+2}(x_3) \\ \vdots & \vdots & \vdots & \vdots \\ b_1(x_N) & b_2(x_N) & \cdots & b_{N+2}(x_N) \\ b_1'(x_N) & b_2''(x_N) & \cdots & b_{N+2}'(x_N) \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \\ \vdots \\ c_N \\ c_{N+1} \\ c_{N+2} \end{pmatrix} = \begin{pmatrix} f'(x_1) \\ f(x_1) \\ f(x_2) \\ f(x_2) \\ \vdots \\ f(x_N) \\ f'(x_N) \end{pmatrix}.$$